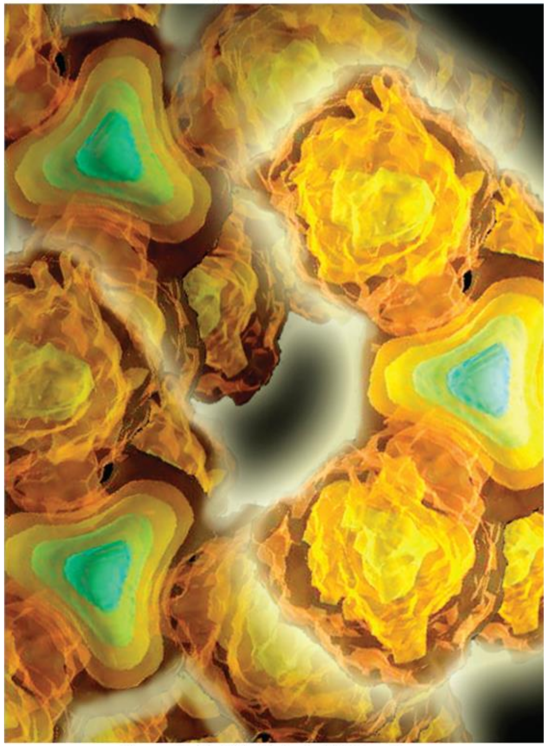
Chapter 7

Quantum Theory and the Electronic Structure of Atoms

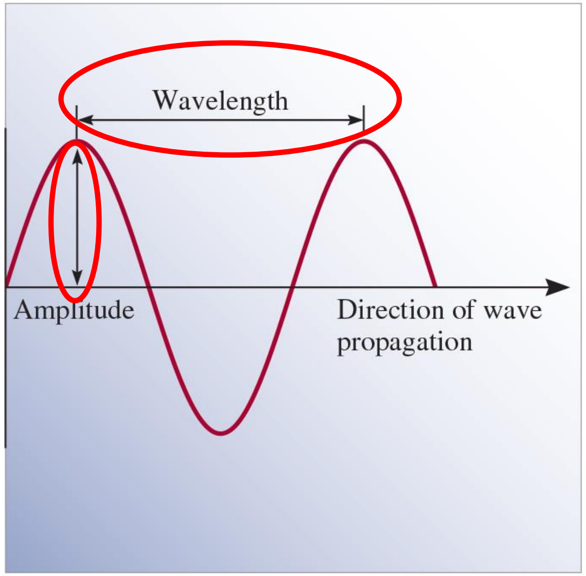


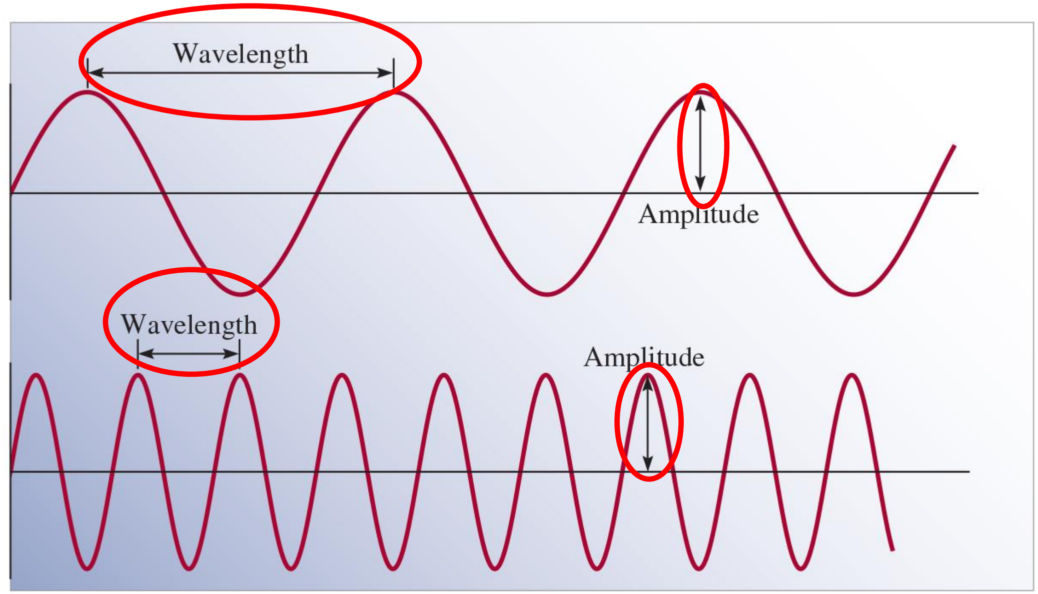
From Classical Physics to Quantum Theory

Max Planck analyzed the data on radiation emitted by solids heated to various temperatures and discovered that atoms and molecules emit energy only in certain discrete quantities, or quanta.

Physicists had always assumed that energy is continuous and that any amount of energy could be released in a radiation process.

Properties of Waves





Wavelength (λ) is the distance between identical points on successive waves.

Amplitude is the vertical distance from the midline of a wave to the peak or trough.

Frequency (ν) is the number of waves that pass through a particular point in 1 second ($\text{Hz} = 1\ \text{cycle}/s$).

The speed of the wave $u = \lambda \nu$

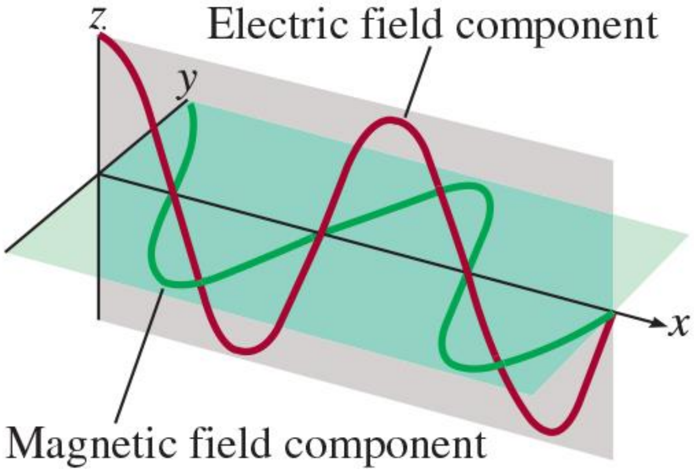
Light as a Wave

Maxwell (1873), proposed that visible light consists of electromagnetic waves which have an electric field component and a magnetic field component. These two components have the same wavelength and frequency, and hence the same speed, but they travel in mutually perpendicular planes.

Electromagnetic radiation is the emission and transmission of energy in the form of electromagnetic waves.

Speed of light (*c*) in vacuum
$= 3.00 \times 10^8\ \mathrm{m/s}$

All electromagnetic radiation
$\lambda\nu = c$



Example 7.1

The wavelength of the green light from a traffic signal is centered at 522 nm. What is the frequency of this radiation?

Solution

Because the speed of light is given in meters per second, it is convenient to first convert wavelength to meters. Recall that $1 \text{ nm} = 1 \times 10^{-9} \text{m}$ (see Table 1.3).

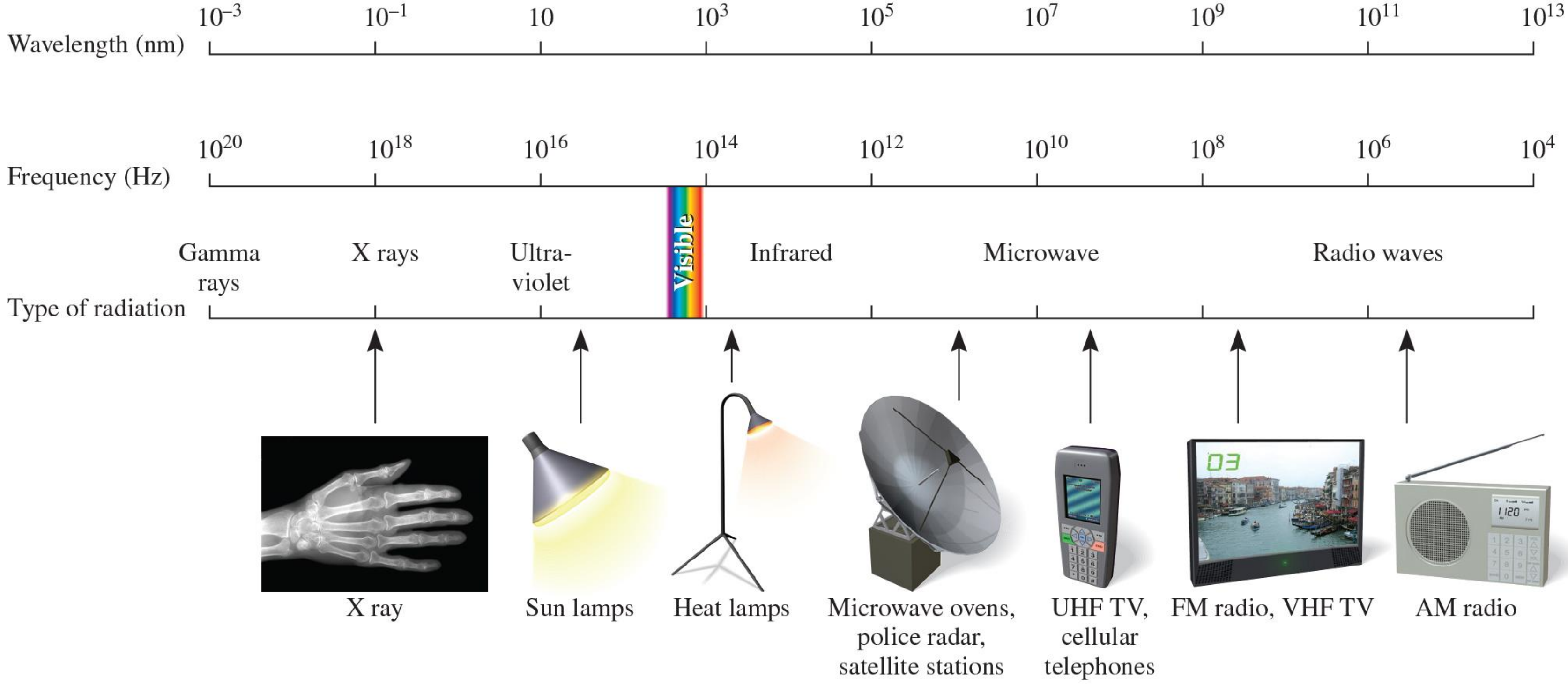
$$v = \frac{c}{\lambda} \Rightarrow \lambda = 522 \text{ nm} \times \frac{1 \times 10^{-9} \text{ m}}{1 \text{ nm}} = 522 \times 10^{-9} \text{ m} = 5.22 \times 10^{-7} \text{ m}$$

Substituting in the wavelength and the speed of light $(3.00 \times 10^{8} \text{ m/s})$, the frequency is

$$v = \frac{3.00 \times 10^{8} \text{ m/s}}{5.22 \times 10^{-7} \text{ m}}$$

$= 5.75 \times 10^{14}/s$, or $5.75 \times 10^{14} \text{ Hz}$

Electromagnetic Spectrum



(a)

(b)

(a) Types of electromagnetic radiation. Gamma rays have the shortest wavelength and highest frequency; radio waves have the longest wavelength and the lowest frequency. Each type of radiation is spread over a specific range of wavelengths (and frequencies). (b) Visible light ranges from a wavelength of 400 nm (violet) to 700 nm (red).

Mystery #1, “Heated Solids Problem” Solved by Planck in 1900

When solids are heated, they emit electromagnetic radiation over a wide range of wavelengths.

<Examples>The dull red glow of an electric heater
The bright white light of a tungsten lightbulb

The amount of radiant energy emitted by an object at a certain temperature depends on its wavelength.

Classical physics assumed that atoms and molecules could emit (or absorb) any arbitrary amount of radiant energy.

Planck said that atoms and molecules could emit (or absorb) energy only in discrete quantities, like small packages or bundles.

Energy (light) is emitted or absorbed in discrete units (quantum).

The energy *E* of a single quantum of energy

$$E = h\nu = h\frac{c}{\lambda}$$

Planck’s constant (*h*)

$$h = 6.63 \times 10^{-34}\ J \cdot s$$

Energy is always emitted in integral multiples of $nh\nu$.

Mystery #2, "Photoelectric Effect" Solved by Einstein in 1905

Light has both:

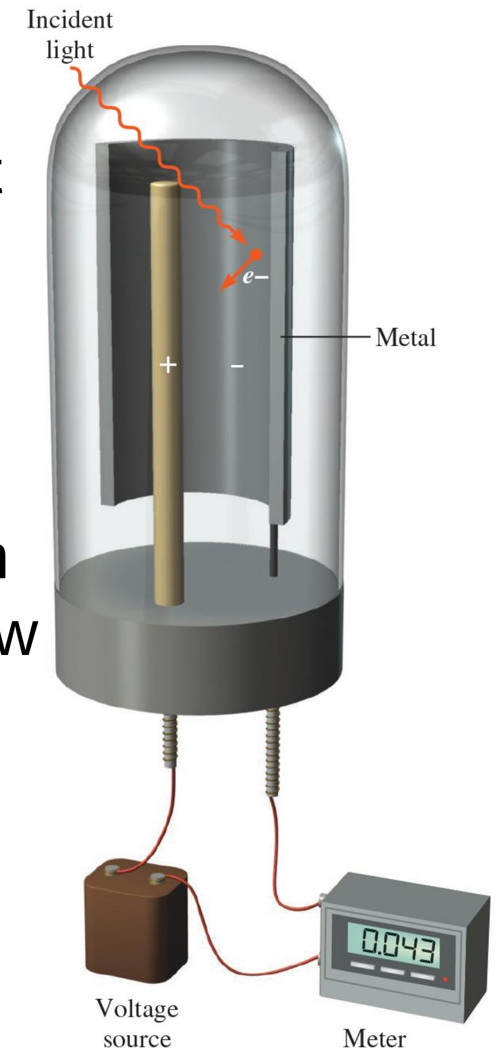
1. wave nature
2. particle nature

Photon is a "particle" of light

$$h\nu = \mathrm{KE} + W$$
$$\mathrm{KE} = h\nu - W$$

KE≡ the kinetic energy of the ejected electron

W ≡ the work function which is a measure how strongly the electrons are held in metal.



Mystery #2, "Photoelectric Effect" Solved by Einstein in 1905 (2)

$$h\nu = \mathrm{KE} + W$$
$$\mathrm{KE} = h\nu - W$$

The more energetic the photon (the higher the frequency), the greater the kinetic energy (KE) of the ejected electron.

Two beams of light having the same frequency (which is greater than the threshold frequency) but different intensities.

⇒The more intense beam of light consists of a larger number of photons; consequently, it ejects more electrons from the metal's surface than the weaker beam of light.
⇒The more intense the light, the greater the number of electrons emitted by the target metal; the higher the frequency of the light, the greater the kinetic energy of the ejected electrons.

Example 7.2

Calculate the energy (in joules) of

(a) a photon with a wavelength of 5.00×10^{4} nm (infrared region)

(b) a photon with a wavelength of 5.00×10^{-2} nm (X ray region)

Solution

(a)From Equation (7.3),

$$E = h\frac{c}{\lambda} = \frac{(6.63 \times 10^{-34}\ \mathrm{J \cdot s})(3.00 \times 10^{8}\ \mathrm{m/s})}{(5.00 \times 10^{4}\ \mathrm{nm})\dfrac{1 \times 10^{-9}\ \mathrm{m}}{1\ \mathrm{nm}}}$$

$$= 3.98 \times 10^{-21}\ \mathrm{J}$$

This is the energy of a single photon with a 5.00×10^{4} nm wavelength.

(b)Following the same procedure as in (a), we can show that the energy of the photon that has a wavelength of 5.00×10^{-2} nm is 3.98×10^{-15} J.

Example 7.3

The work function of cesium metal is 3.42×10^{-19} J.

(a)Calculate the minimum frequency of light required to release electrons from the metal.

(b)Calculate the kinetic energy of the ejected electron if light of frequency $1.00 \times 10^{15} s^{-1}$ is used for irradiating the metal.

Solution

(a) Setting $\mathrm{KE} = 0$ in Equation (7.4), we write $h\nu = W$

Thus,
$$\nu = \frac{W}{h} = \frac{3.42 \times 10^{-19}\ \mathrm{J}}{6.63 \times 10^{-34}\ \mathrm{J \cdot s}}$$
$$= 5.16 \times 10^{14}\ \mathrm{s^{-1}}$$

(b) Rearranging Equation (7.4) gives

$$\begin{aligned}\mathrm{KE} &= h\nu - W \\ &= (6.63 \times 10^{-34}\ \mathrm{J \cdot s})(1.00 \times 10^{15}\ \mathrm{s^{-1}}) - 3.42 \times 10^{-19}\ \mathrm{J} \\ &= 3.21 \times 10^{-19}\ \mathrm{J}\end{aligned}$$

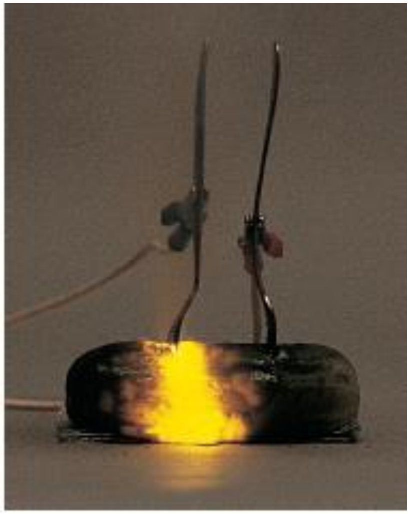
Bohr's Theory of the Hydrogen Atom

Emission Spectra

The emission spectrum of a substance can be seen by energizing a sample of material either with thermal energy or with some other form of energy (such as a high-voltage electrical discharge).

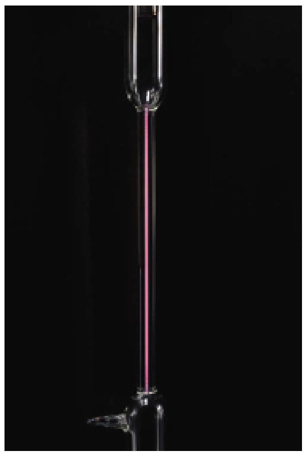
<Example>

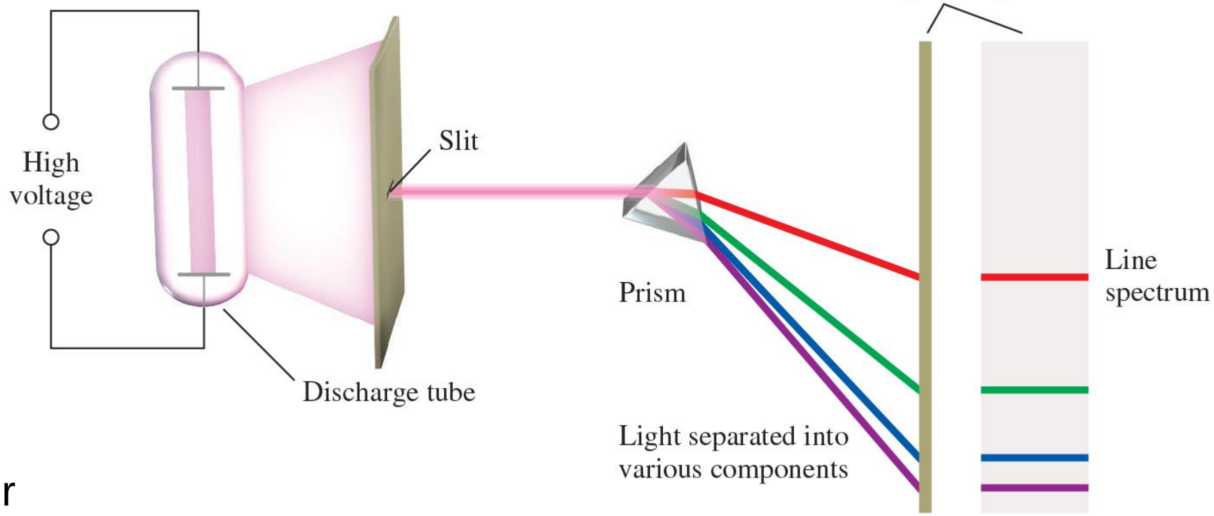
1. A "red-hot" or "white-hot" iron bar freshly removed from a high-temperature source produces a characteristic glow.
2. The warmth of the same iron bar represents another portion of its emission spectrum—the infrared region.
3. When a high voltage is applied between the forks, some of the sodium ions in the pickle are converted to sodium atoms in an excited state. These atoms emit the characteristic yellow light as they relax to the ground state.



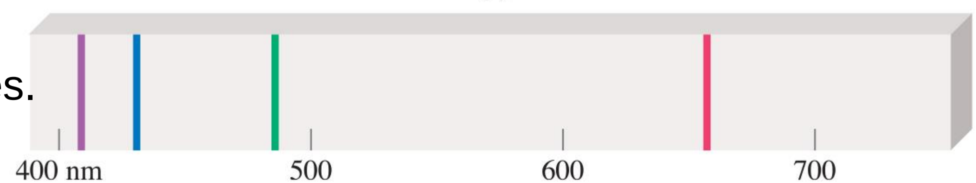
Line Emission Spectrum of Hydrogen Atoms (1)

The emission spectra of atoms in the gas phase, on the other hand, do not show a continuous spread of wavelengths from red to violet; rather, the atoms produce bright lines in different parts of the visible spectrum. These line spectra are the light emission only at specific wavelengths.





(a)

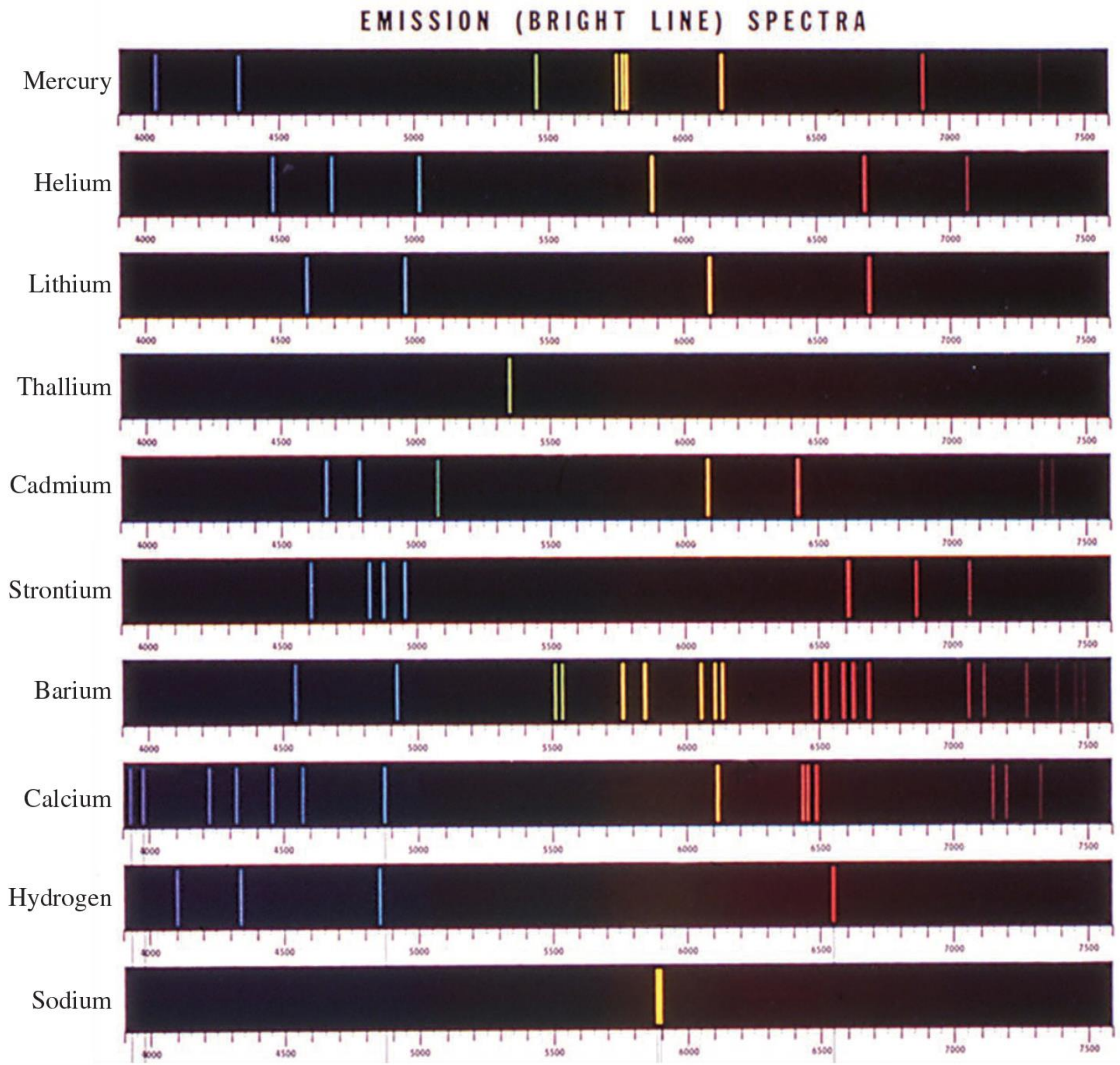


(b)

(a) An experimental arrangement for studying the emission spectra of atoms and molecules. The gas under study is in a discharge tube containing two electrodes. As electrons flow from the negative electrode to the positive electrode, they collide with the gas. This collision process eventually leads to the emission of light by the atoms (or molecules). The emitted light is separated into its components by a prism. Each component color is focused at a definite position, according to its wavelength, and forms a colored image of the slit on the photographic plate. The colored images are called spectral lines.
(b) The line emission spectrum of hydrogen atoms.

Emission Spectra of Some Elements

The characteristic lines in atomic spectra can be used in chemical analysis to identify unknown atoms.



Emission Spectrum of the Hydrogen Atom

1. e^- can only have specific (quantized) energy values
2. light is emitted as e^- moves from one energy level to a lower energy level

$$E_n = -R_H\left(\frac{1}{n^2}\right)$$

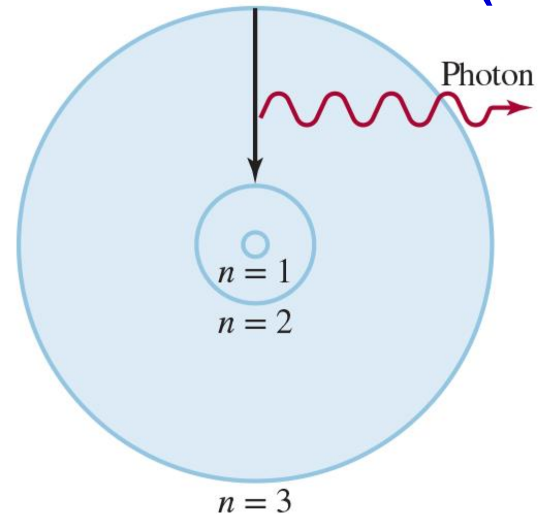
The energies that an electron in a hydrogen atom

n (principal quantum number) $= 1,2,3, \ldots$

R_H (Rydberg constant) $= 2.18 \times 10^{-18}\text{J}$

The negative sign in Equation is an arbitrary convention, signifying that the energy of the electron in the atom is lower than the energy of a **free electron**, which is an electron that is infinitely far from the nucleus.

Bohr's Model of the Atom (1913)



The emission process in an excited hydrogen atom, according to Bohr's theory. An electron originally in a higher energy orbit (n=3) falls back to a lower-energy orbit (n=2). As a result, a photon with energy $h\nu$ is given off. The value of $h\nu$ is equal to the difference in energies of the two orbits occupied by the electron in the emission process.

Emission Spectrum of the Hydrogen Atom (1)

The energy of a free electron is arbitrarily assigned a value of zero. $E_n = -R_{\mathrm{H}}\left(\frac{1}{n^2}\right) \Rightarrow E_\infty \to 0$

$$E_n = -R_{\mathrm{H}}\left(\frac{1}{n^2}\right)$$

$$\begin{cases} n = 1, \text{ground state (level)— the lowest energy state of a system} \\ n \geq 2, \text{excited state (level)−−higher in energy than the ground state} \end{cases}$$

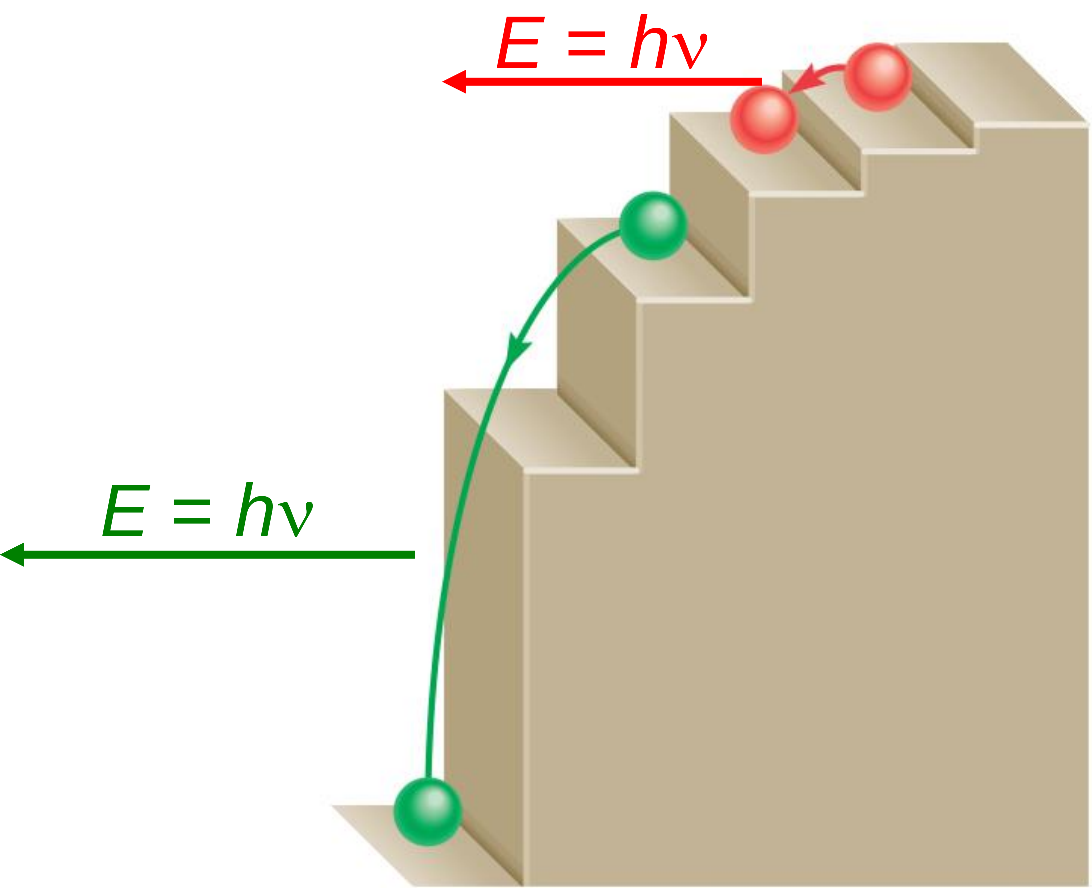
As the electron gets closer to the nucleus (as n decreases), E_n becomes larger in absolute value, but also more negative.

The radius of each circular orbit in Bohr's model depends on n^2.

As n increases from 1 to 2 to 3, the orbit radius increases very rapidly. The higher the excited state, the farther away the electron is from the nucleus (and the less tightly it is held by the nucleus).

Emission Spectrum of the Hydrogen Atom (1)

Quantized Energy



The quantized movement of the electron from one energy state to another is analogous to the movement of a tennis ball either up or down a set of stairs.
The ball can be on any of several steps but never between steps.

Energy Transitions of the Hydrogen Atom

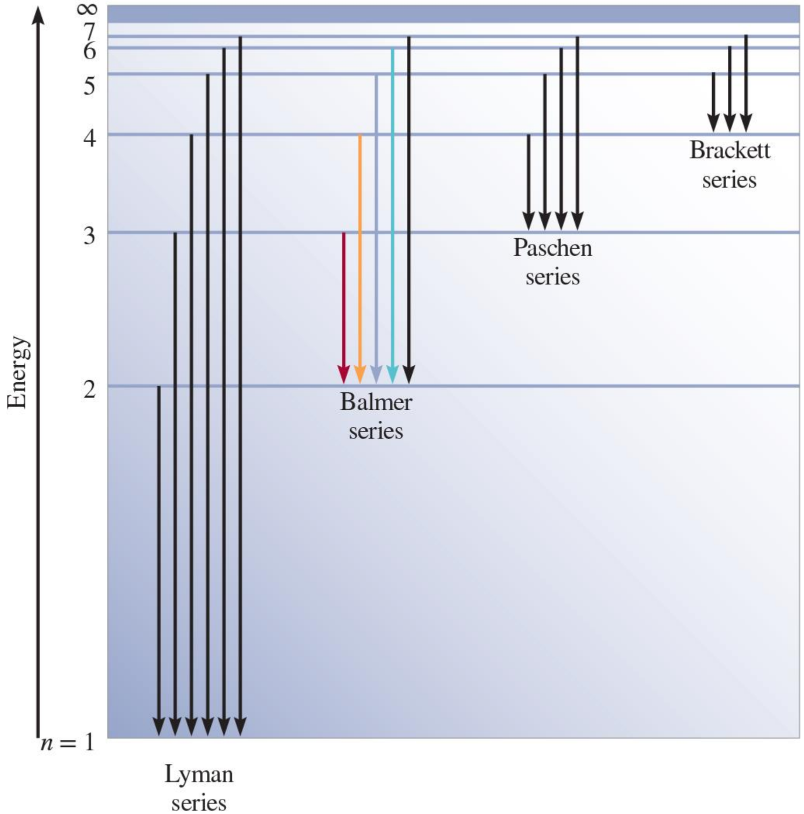


Table 7.1 The Various Series in Atomic Hydrogen Emission Spectrum

Series	n_f	n_i	Spectrum Region
Lyman	1	2,3,4,…	Ultraviolet
Balmer	2	3,4,5,…	Visible and ultraviolet
Paschen	3	4,5,6,…	Infrared
Brackett	4	5,6,7,…	Infrared

$$E_n = -R_{\mathrm{H}}\left(\frac{1}{n^2}\right)$$

$$E_{\text{photon}} = \Delta E = E_f - E_i$$

$$E_{\mathrm{i}} = -R_{\mathrm{H}}\left(\frac{1}{n_i^2}\right) \qquad E_f = -R_{\mathrm{H}}\left(\frac{1}{n_f^2}\right)$$

$$\Delta E = h\nu = R_{\mathrm{H}}\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

Example 7.4

What is the wavelength of a photon (in nanometers) emitted during a transition from the $n_\mathrm{i} = 5$ state to the $n_\mathrm{f} = 2$ state in the hydrogen atom?

Solution From Equation (7.6) we write

$$\Delta \mathrm{E} = R_\mathrm{H}\left(\frac{1}{n_\mathrm{i}^2} - \frac{1}{n_\mathrm{f}^2}\right) = 2.18 \times 10^{-18}\ \mathrm{J}\left(\frac{1}{5^2} - \frac{1}{2^2}\right) = -4.58 \times 10^{-19}\ \mathrm{J}$$

The negative sign indicates that this is energy associated with an emission process. To calculate the wavelength, we will omit the minus sign for ΔE because the wavelength of the photon must be positive.

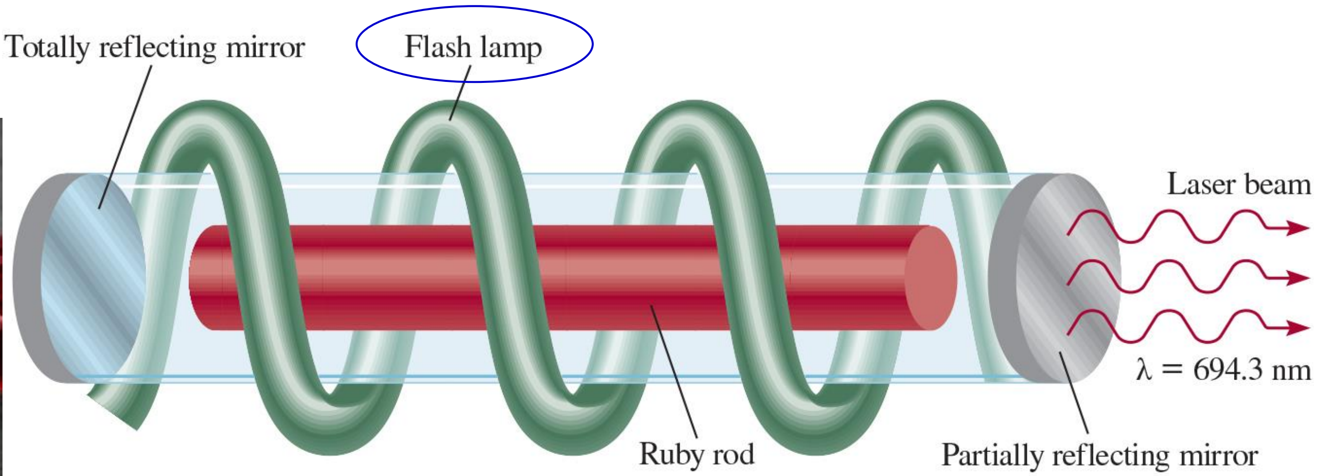
Because $\Delta E = h\mathrm{v}$ or $\mathrm{v} = \Delta E/h$, we can calculate the wavelength of the photon by writing

$$\lambda = \frac{c}{v} = \frac{ch}{\Delta E} = \frac{(3.00 \times 10^{8}\ \mathrm{m/s})(6.63 \times 10^{-34}\ \mathrm{J \cdot s})}{4.58 \times 10^{-19}\ \mathrm{J}} = 4.34 \times 10^{-7}\ \mathrm{m}$$

$$= 4.34 \times 10^{-7}\ \mathrm{m} \times \left(\frac{1\ \mathrm{nm}}{1 \times 10^{-9}\ \mathrm{m}}\right) = 434\ \mathrm{nm}$$

Chemistry in Action: Laser – The Splendid Light

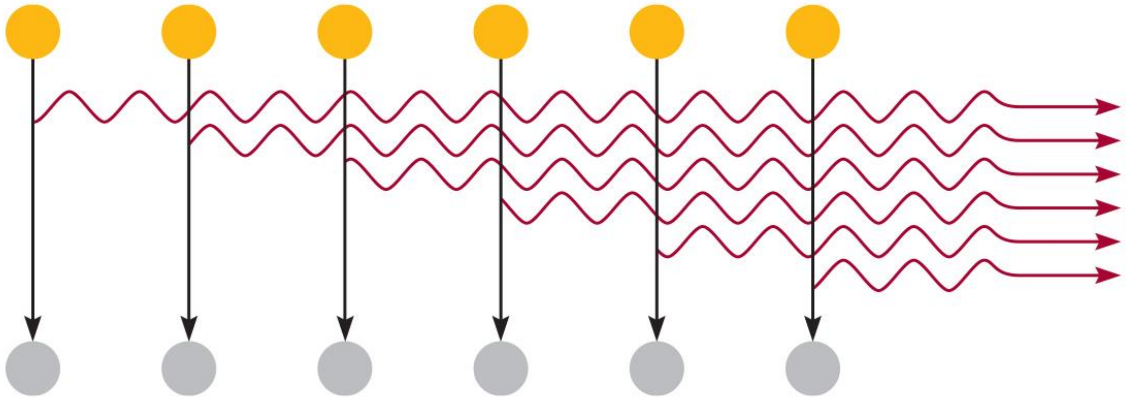
Laser is an acronym for **l**ight **a**mplification by **s**timulated **e**mission of **r**adiation.



Ruby is a deep-red mineral containing corundum, Al_2O_3, in which some of the Al^{3+} ions have been replaced by Cr^{3+} ions.

A flashlamp is used to excite the chromium atoms to a higher energy level. The excited atoms are unstable, so at a given instant some of them will return to the ground state by emitting a photon in the red region of the spectrum.

Chemistry in Action: Laser – The Splendid Light (1)



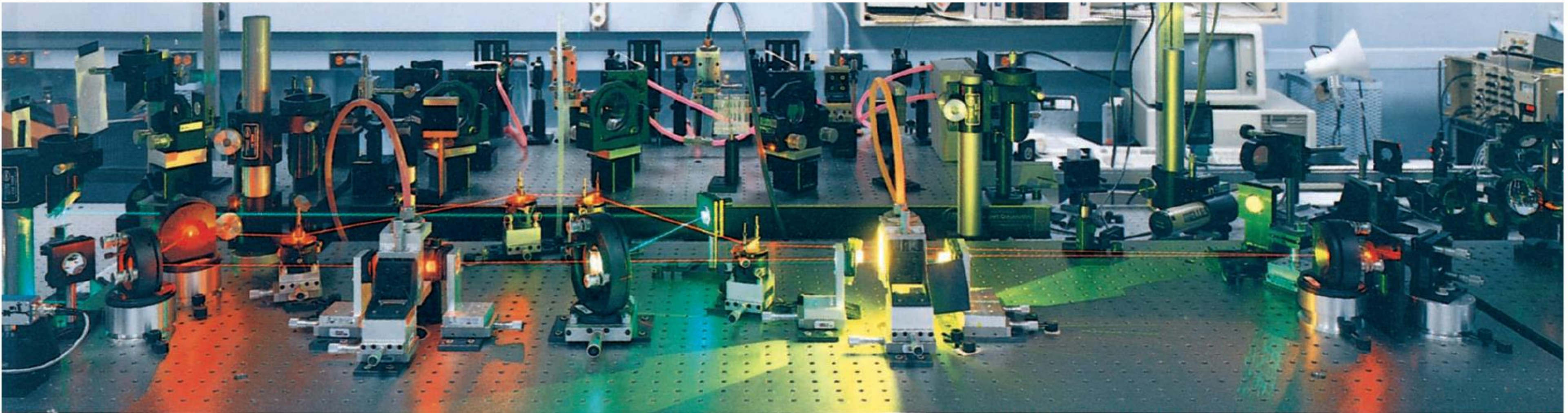
The photon bounces back and forth many times between mirrors at opposite ends of the laser tube. This photon can stimulate the emission of photons of exactly the same wavelength from other excited chromium atoms; these photons in turn can stimulate the emission of more photons, and so on.

Because the light waves are in phase—that is, their maxima and minima coincide—the photons enhance one another, increasing their power with each passage between the mirrors. One of the mirrors is only partially reflecting, so that when the light reaches a certain intensity it emerges from the mirror as a laser beam.
The laser light may be emitted in pulses (ruby laser) or in continuous waves.

Chemistry in Action: Laser – The Splendid Light (2)

Laser light is
(1) intense–precisely known wavelength and hence energy
(2) monoenergetic,
(3) coherent–the light waves are all in phase.

Applications–eye surgery, for drilling holes in metals and welding, and for carrying out nuclear fusion.

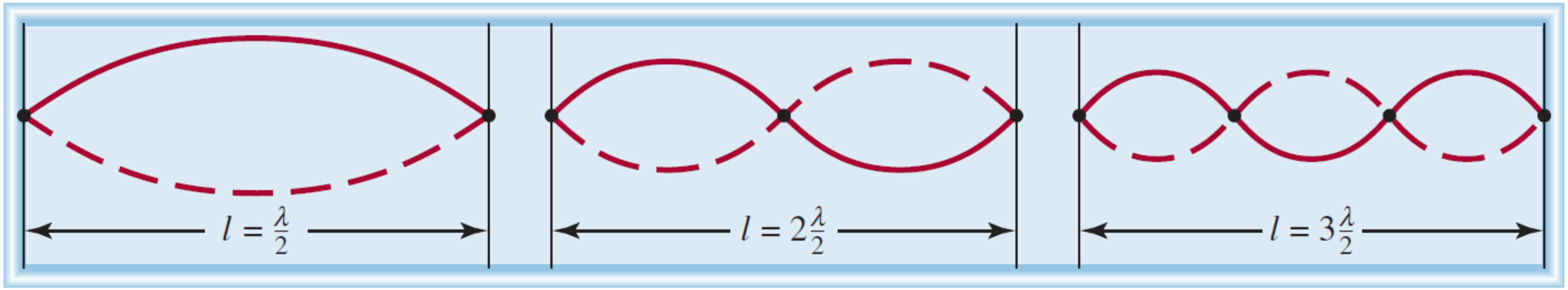


Why is the electron in a Bohr atom restricted to orbiting the nucleus at certain fixed distances?

For a decade no one, not even Bohr himself, had a logical explanation.

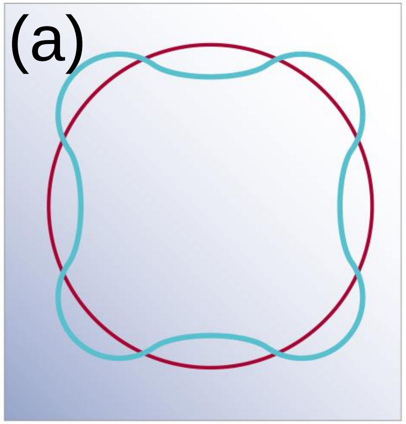
De Broglie (1924)–if light waves can behave like a stream of particles (photons), then perhaps particles such as electrons can possess wave properties.

⇒An electron bound to the nucleus behaves like a standing wave. The waves are described as standing, or stationary–they do not travel along the string. Some points (nodes) on the string do not move at all–the amplitude of the wave at these points is zero.



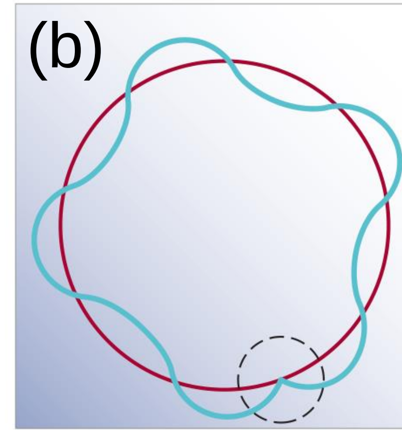
The standing waves generated by plucking a guitar string. Each dot represents a node. The length of the string (l) must be equal to a whole number times one-half the wavelength ($\lambda/2$).

The Dual Nature of the Electron–Quantization of Electron Energy



(a) The circumference of the orbit is equal to an integral number of wavelengths.
⇒An allowed orbit.

(b) The circumference of the orbit is not equal to an integral number of wavelengths. →the electron wave does not close in on itself. ⇒A nonallowed orbit. The wave would partially cancel itself on each successive orbit. Eventually the amplitude of the wave would be reduced to zero, and the wave would not exist.



De Broglie (1924) reasoned that e^- is both particle and wave.

$$2\pi r = n\lambda;\ n \in \mathbb{Z}^+, \quad \lambda = \frac{h}{mu}$$

λ = wavelength of particle (e^-);
u = velocity of particle (e^-);
m = mass of particle (e^-)

Waves can behave like particles and particles can exhibit wavelike properties.

Example 7.5

Calculate the wavelength of the "particle" in the following two cases:

(a) The fastest serve in tennis is about 150 miles per hour, or $68\ \mathrm{m/s}$. Calculate the wavelength associated with a $6.0 \times 10^{-2} - \mathrm{kg}$ tennis ball traveling at this speed.

(b) Calculate the wavelength associated with an electron $(9.1094 \times 10^{-31}\ \mathrm{kg})$ moving at $68\ \mathrm{m/s}$.

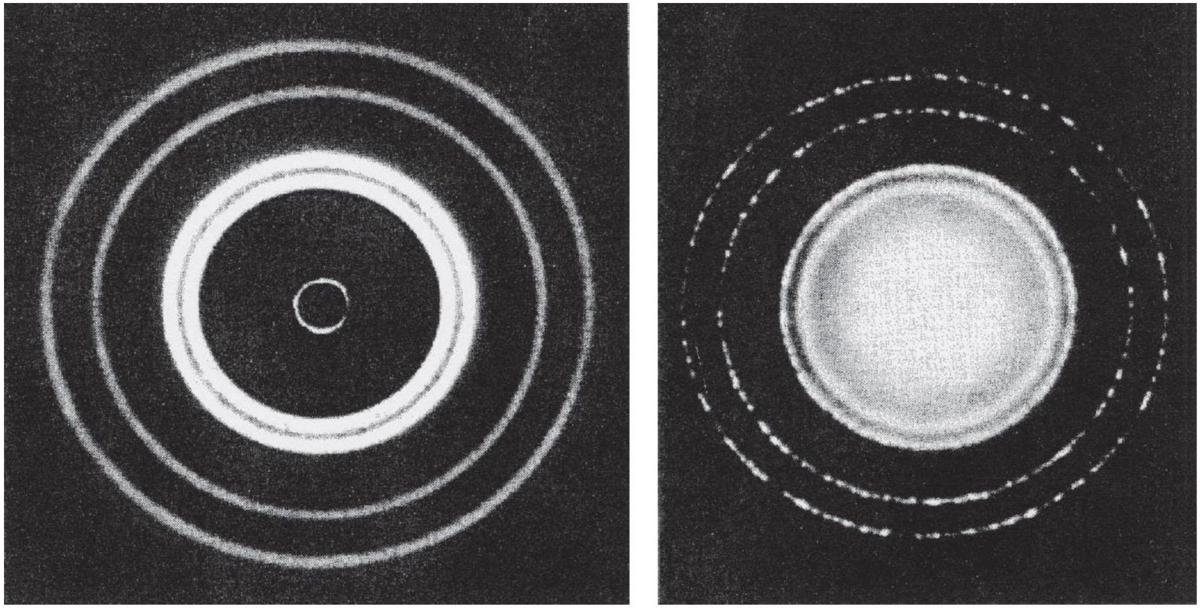
Solution

$$(\mathrm{a})\lambda = \frac{h}{mu} = \frac{6.63 \times 10^{-34}\ \mathrm{J \cdot s}}{(6.0 \times 10^{-2}\ \mathrm{kg}) \times 68\ \mathrm{m/s}} = 1.6 \times 10^{-34}\ \mathrm{m}$$

Comment This is an exceedingly small wavelength considering that the size of an atom itself is on the order of $1 \times 10^{-10}\mathrm{m}$ ⇒The wave properties of a tennis ball cannot be detected by any existing measuring device.

$$(b)\lambda = \frac{h}{mu} = \frac{6.63 \times 10^{-34}\ \mathrm{J \cdot s}}{(9.1094 \times 10^{-31}\ \mathrm{kg}) \times 68\ \mathrm{m/s}} = 1.1 \times 10^{-5}\ \mathrm{m}$$

Comment This wavelength $(1.1 \times 10^{-5}\ \mathrm{m\ or\ } 1.1 \times 10^{4}\ \mathrm{nm})$ is in the infrared region. This calculation shows that only electrons (and other submicroscopic particles) have measurable wavelengths.

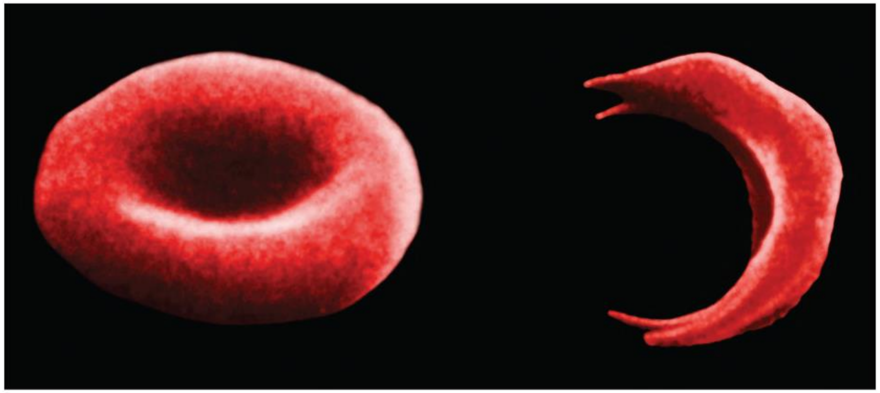


(a) (b)

(a) X-ray (waves) diffraction pattern of aluminum foil. (b) Electron diffraction of aluminum foil. The similarity of these two patterns shows that electrons can behave like X rays and display wave properties.

Chemistry in Action: Electron Microscopy

Electron micrograph of a normal red blood cell and a sickled red blood cell from the same person

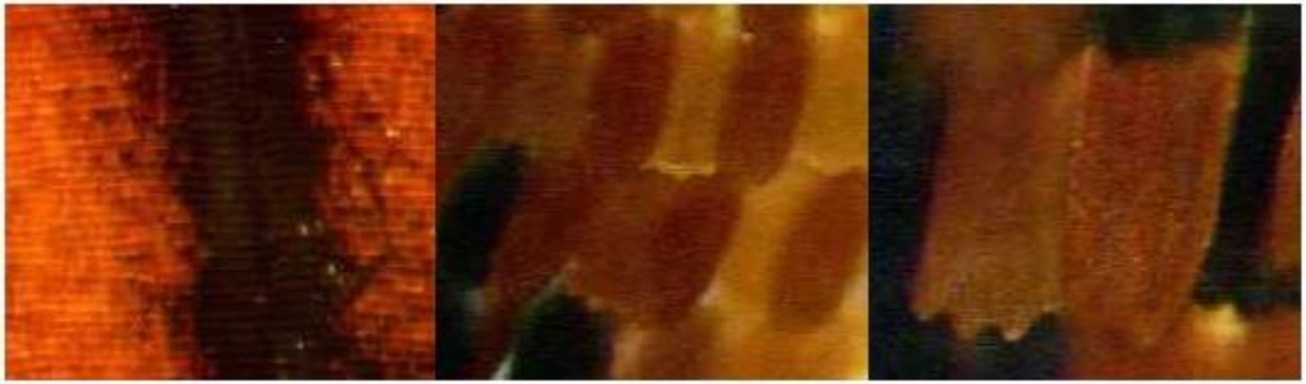


Because the range of visible light wavelengths starts at ~ 400 nm, or 4×10^{-5} cm, we cannot see anything smaller than 2×10^{-5} cm.
We can see objects on the atomic and molecular scale by using X rays, whose wavelengths range from about 0.01 nm to 10 nm.
X rays cannot be focused⇒do not produce well-formed images.
Electrons are charged particles, which can be focused in the same way the image on a TV screen is focused–by applying an electric field or a magnetic field.

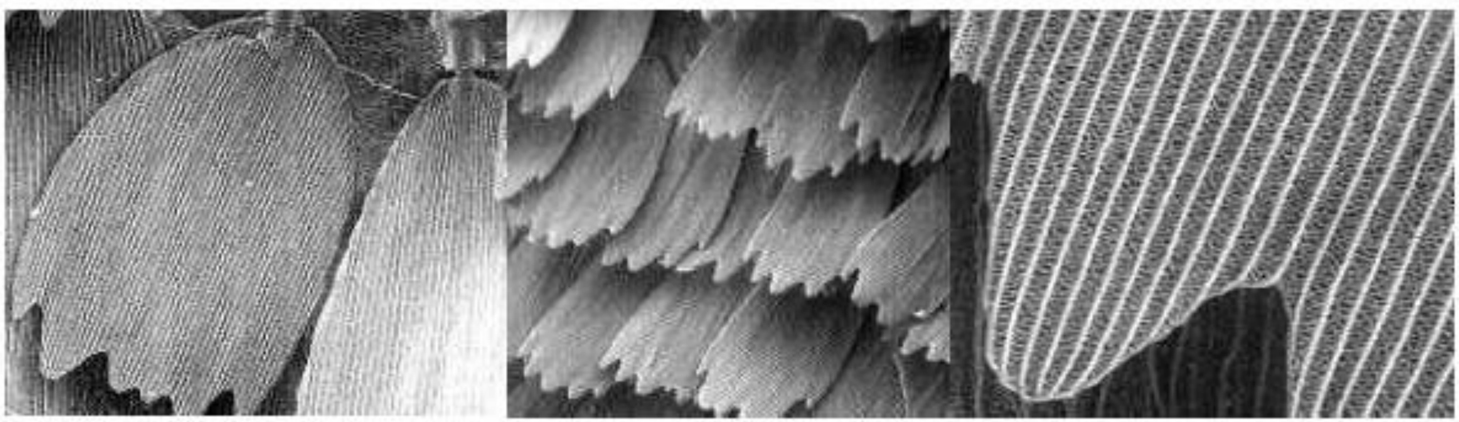
$$\lambda = \frac{h}{mu}$$

By accelerating electrons to very high velocities, we can obtain wavelengths as short as 0.004 nm. $\lambda_e = 0.004\ \text{nm}$

蝴蝶翅膀

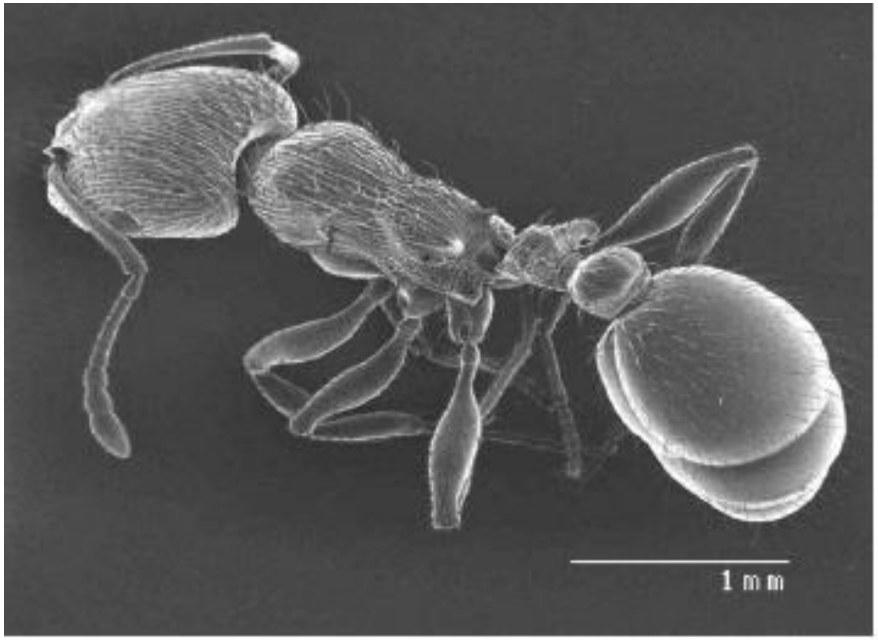


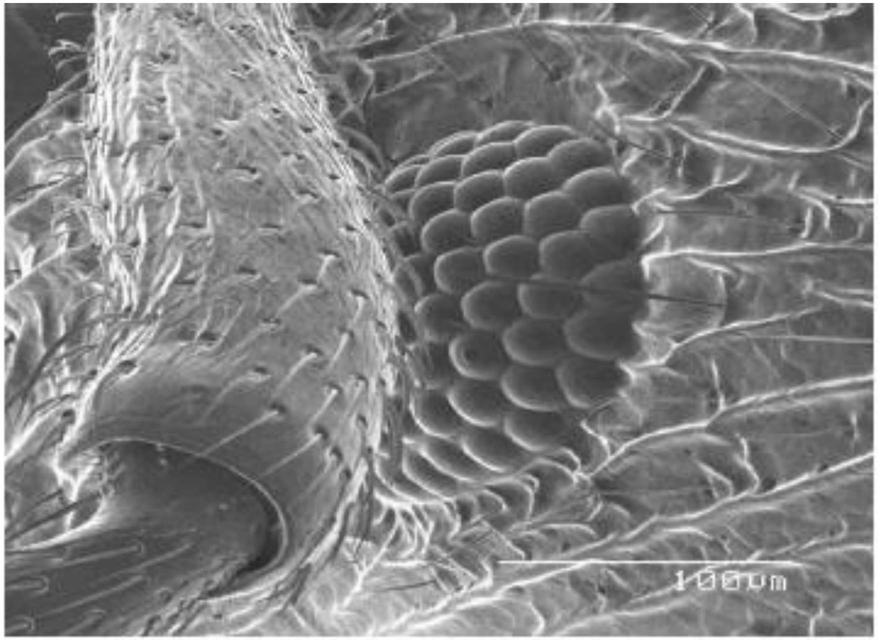
OM



SEM

螞蟻和它的眼睛





Chemistry in Action: Electron Microscopy (1)

The scanning tunneling microscope (STM) makes use of quantum mechanical property of the electron to produce an image of the atoms on the surface of a sample.

Because of its extremely small mass, an electron is able to move or "tunnel" through an energy barrier (instead of going over it).

The STM consists of a tungsten metal needle with a very fine point, the source of the tunneling electrons.

A voltage is maintained between the needle and the surface of the sample to induce electrons to tunnel through space to the sample.

As the needle moves over the sample, at a distance of a few atomic diameters from the surface, the tunneling current is measured. This current decreases with increasing distance from the sample. By using a feedback loop, the vertical position of the tip can be adjusted to a constant distance from the surface. The extent of these adjustments, which profile the sample, is recorded and displayed as a three- dimensional false-colored image.

奈米材料結構檢測的 STM 和 AFM 技術

STM 掃描穿隧顯微鏡 (scanning tunneling microscopy)，圖3-28 穿隧效應示意圖即可以直接觀察到物質表面的原子結構。另外，STM 還具有搬動原子的功能
STM 的工作原理是建立在量子力學中的穿隧效應基礎之上的。
古典物理學:當一個粒子的動能 E 低於前方勢壘的高度 Φ_0 時，它是不能夠穿越此勢壘的，即透射係數等於零；量子力學的計算結果:在一般情況下，其透射係數不等於零，也就是說存在粒子穿越比它能量更高勢壘的可能，即可以產生穿隧電流，如圖 3-28 所示，這個現象稱為穿隧效應。

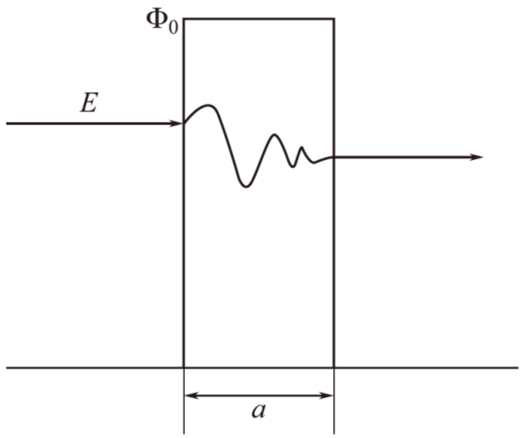
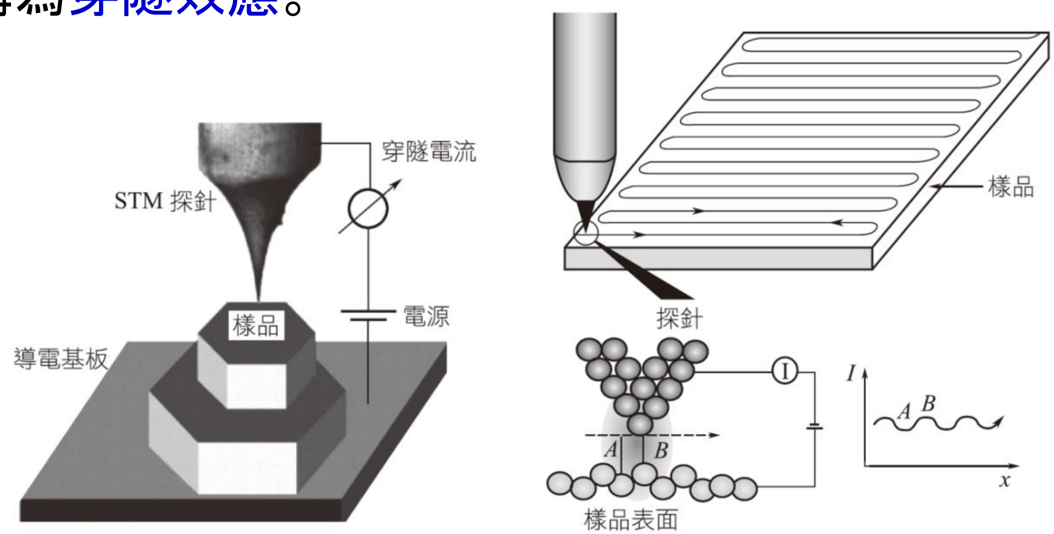


圖 3-28 穿隧效應示意圖



(a) STM 裝置示意圖

(b) 樣品掃描方式及穿隧電流的變化規律

圖 3-29 STM 裝置示意圖和穿隧電流的產生及變化規律

穿隧電流強度同探針針尖和樣品之間的距離 [圖 3-28 中 *a* 值，圖 3-29(b) 中 A、B 值] 有著指數變化關係，如果距離減小 0.1 nm，穿隧電流約可增加一個數量級。
穿隧電流的變化反映的是樣品表面細微的高低起伏變化，將穿隧電流放大並轉化成圖像信號，就可以得到樣品表面的結構資訊⇒STM 的工作原理

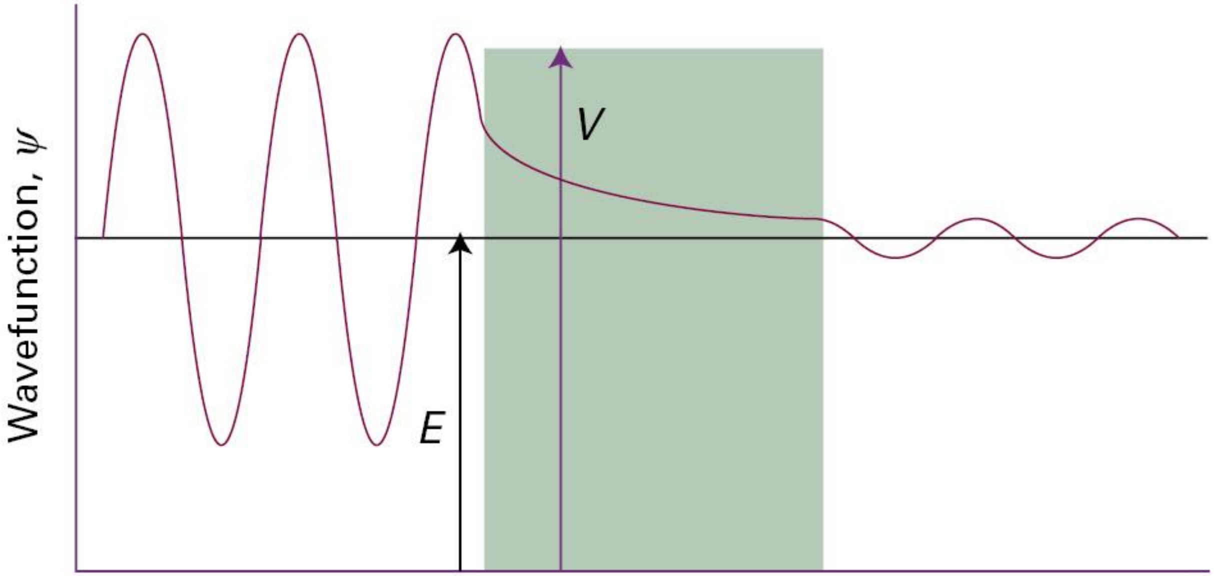


Figure 8A.9 A particle incident on a barrier from the left has an oscillating wave function, but inside the barrier there are no oscillations (for $E < V$). If the barrier is not too thick, the wavefunction is nonzero at its opposite face, and so oscillates begin again there. (Only the real component of the wavefunction is shown.)

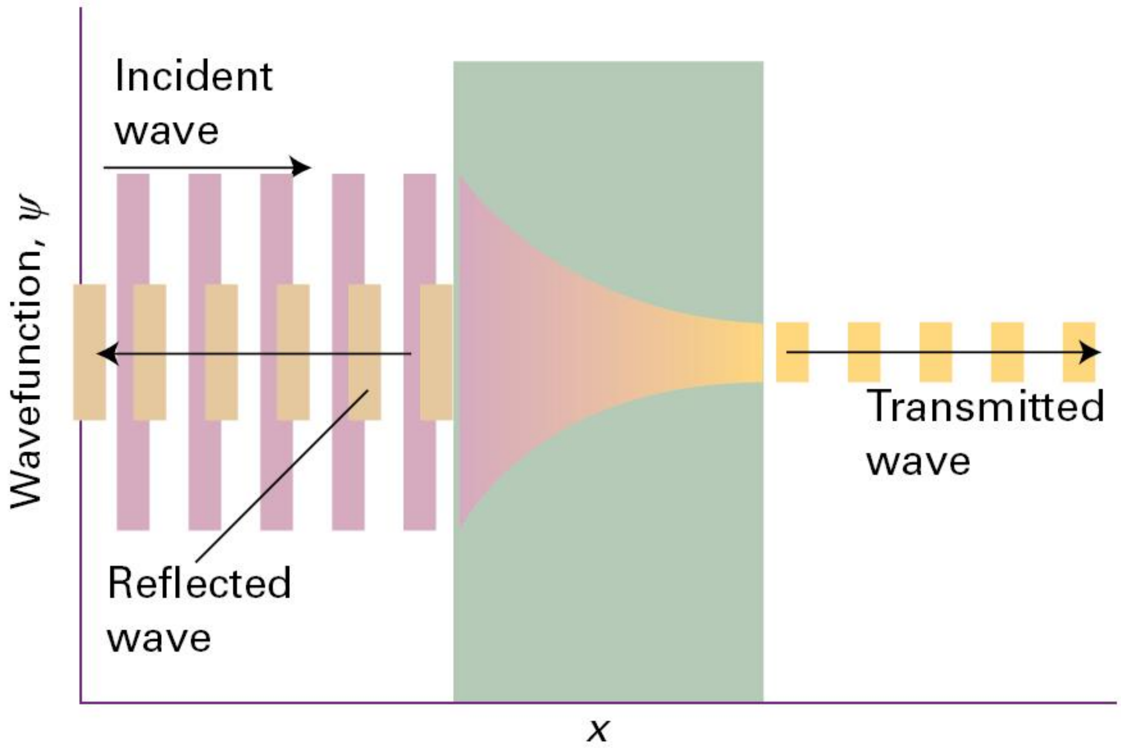
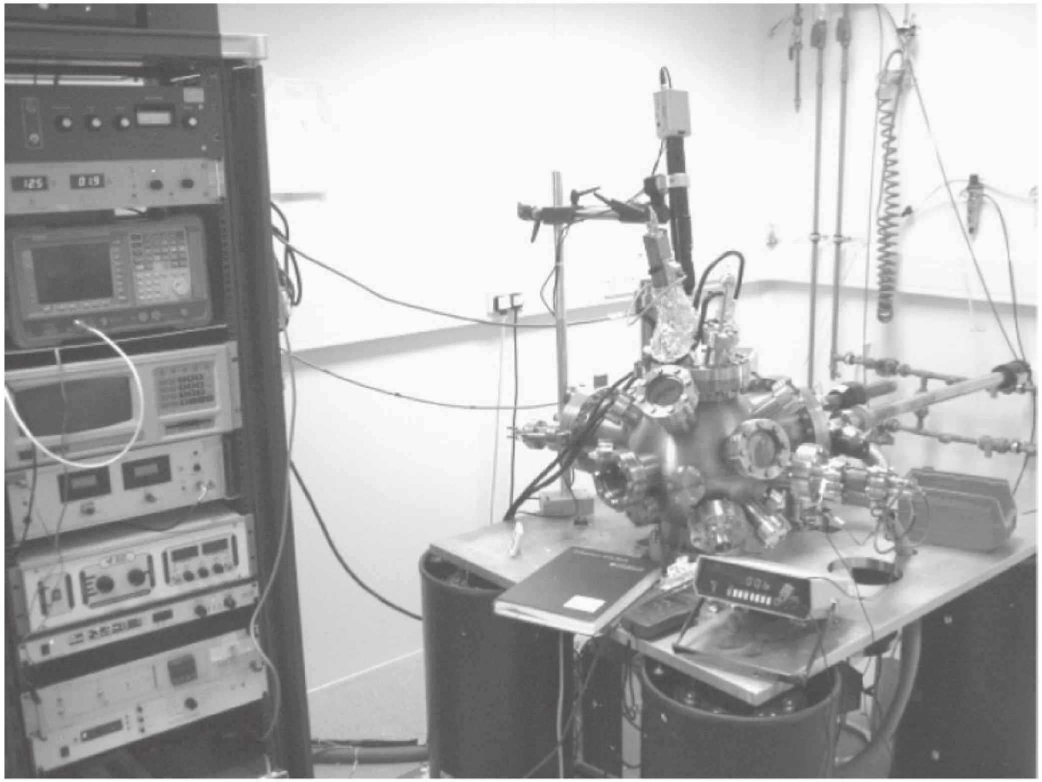
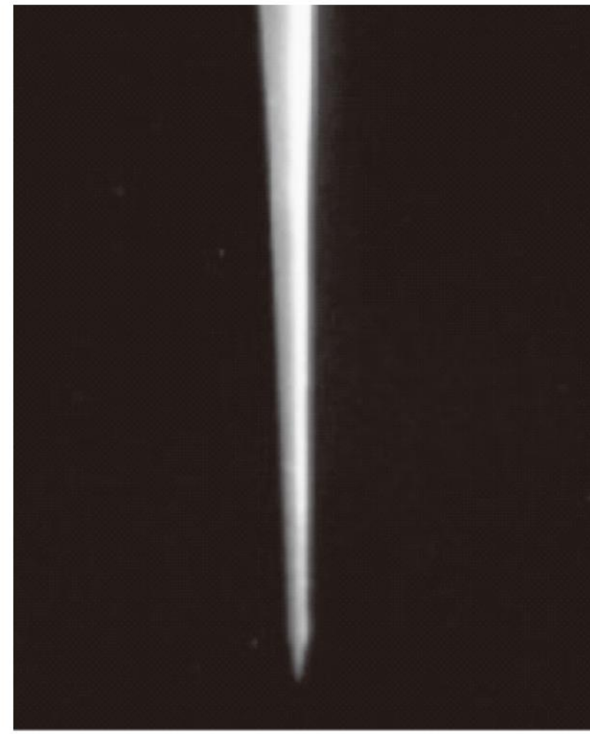


Figure 8A.10 When a particle is incident on a barrier from the left, the wavefunction consists of a wave representing linear momentum to the right, a reflected component representing momentum to the left, a varying but not oscillating component inside the barrier, and a (weak) wave representing motion to the right on the far side of the barrier.



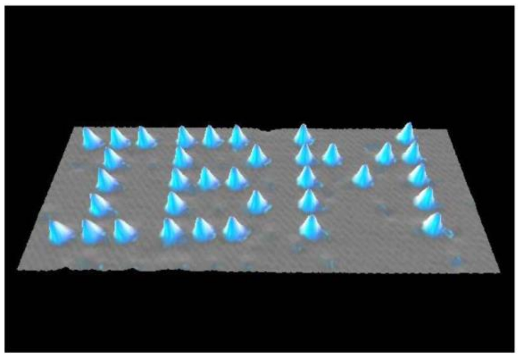
(a) STM 的裝置圖



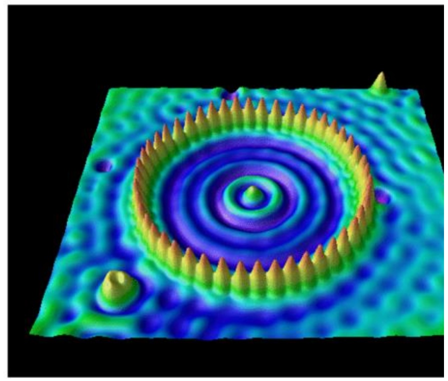
(b) 探針的放大圖像

圖 3-30 STM 的裝置圖和探針的放大圖像

在 Ni 晶體的表面，使用 35 個 Xe 原子「寫」出了 IBM 這 3 個字母，設計了當時世界上最小的人工圖案。不僅如此，這項 STM 技術還在資訊高密度存儲領域有著潛在應用價值。



Xe atoms on Ni(110) are arranged into the letters “IBM”



A quantum corral made of 48 F

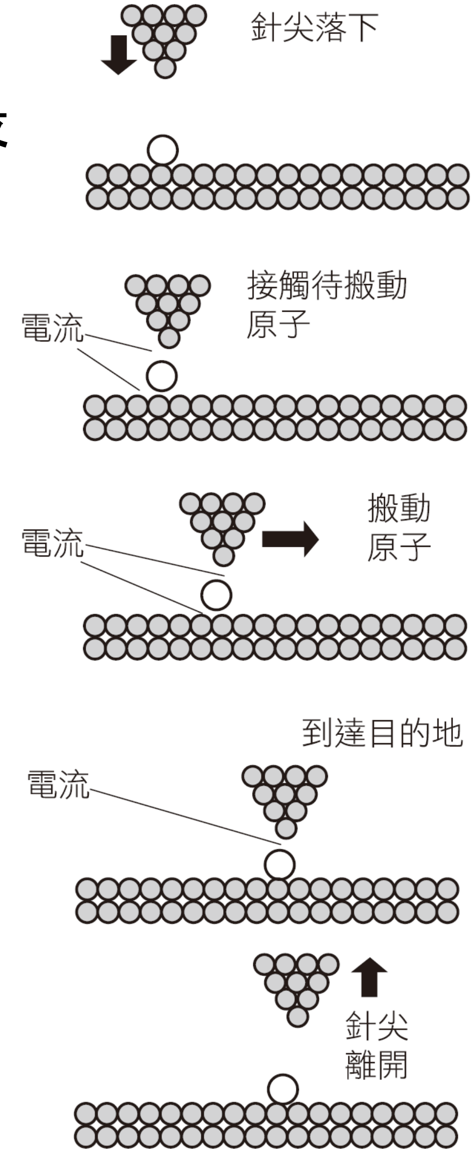
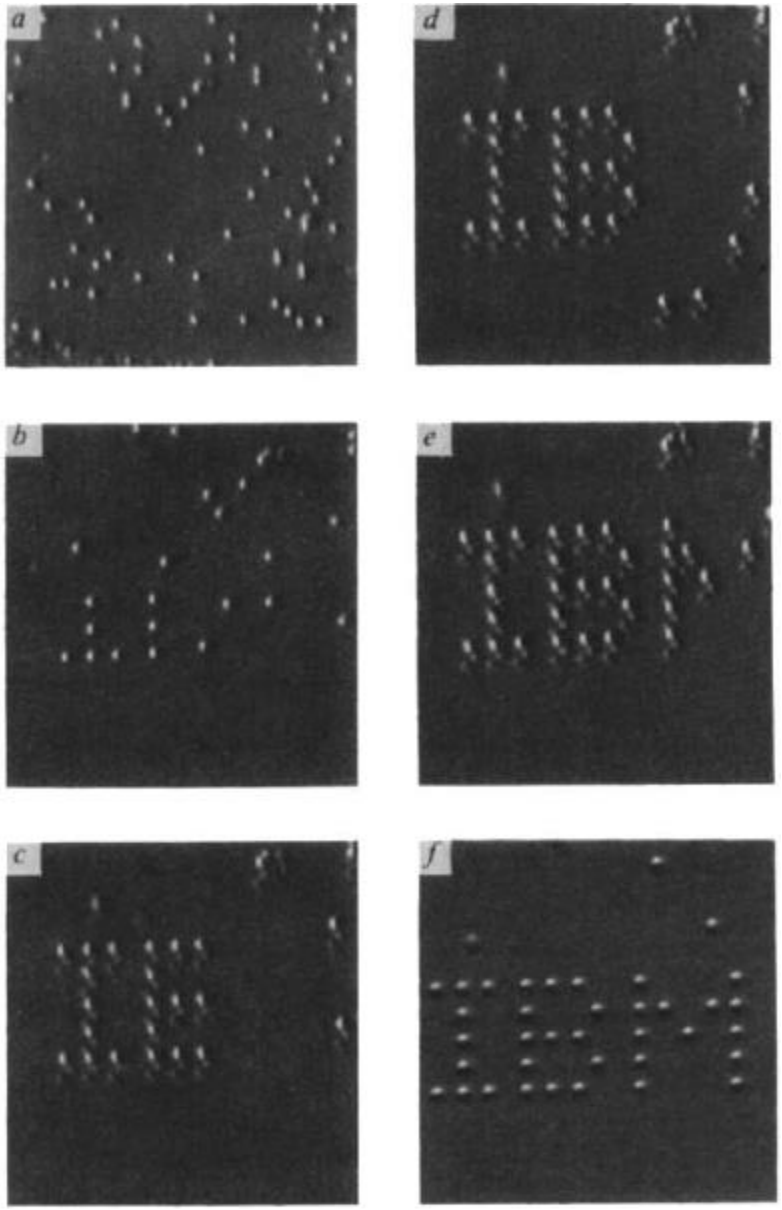
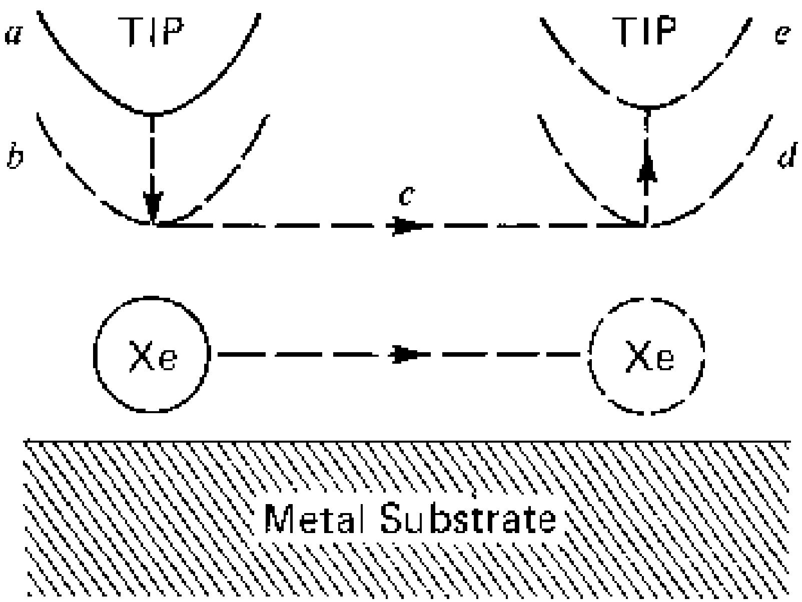


圖 3-33 STM 技術搬運原子的過程示意圖



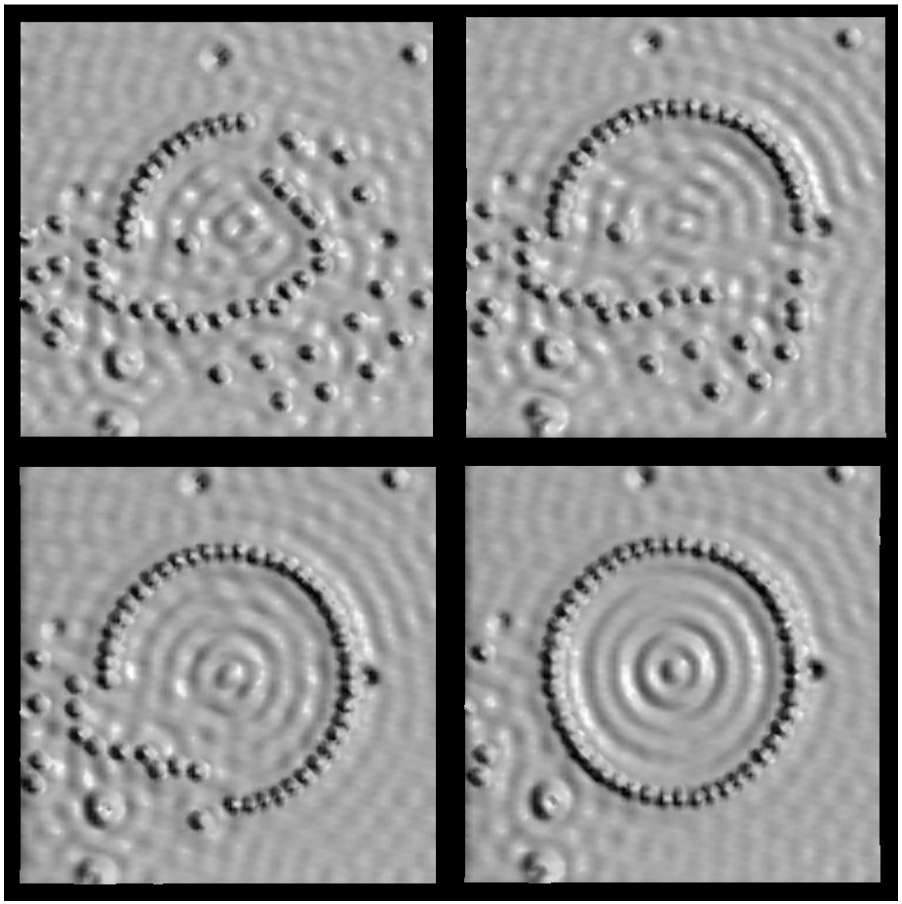
Xe on Ni(110)

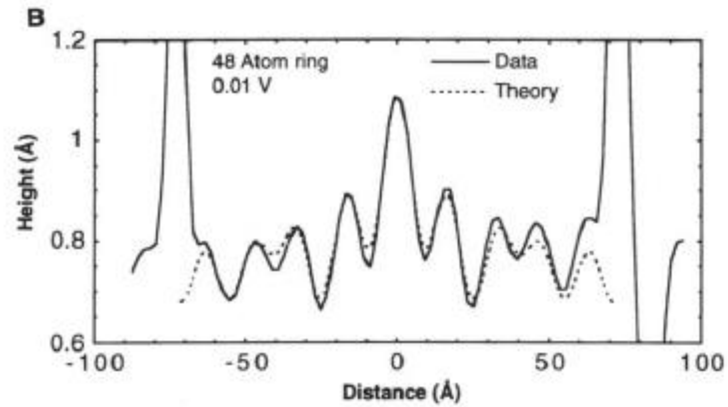
4.5 K UHV STM



D. Eigler et al, Nature **344**, 524 (1990)

Build a quantum corral



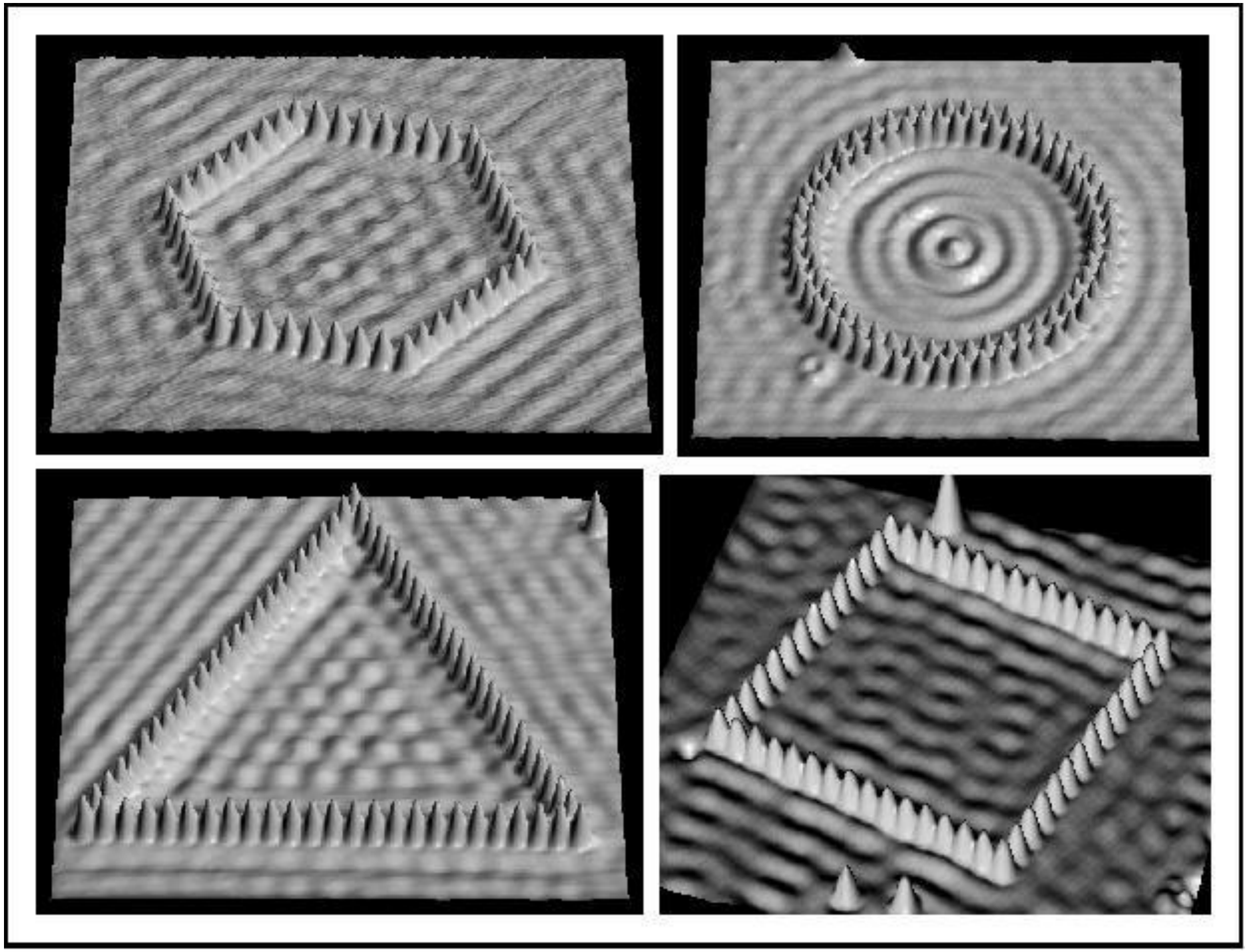


Exp.-theory: **B**) Solid line: cross section of the above data. Dashed line: fit to cross section using a linear combination of |5,0⟩, |4,2⟩, and |2,7⟩ eigenstate densities.

Fe on Cu(111)
4.5K UHV STM

M. Crommie et al, Science **262**, 218 (1993)

What can you do with a corral?

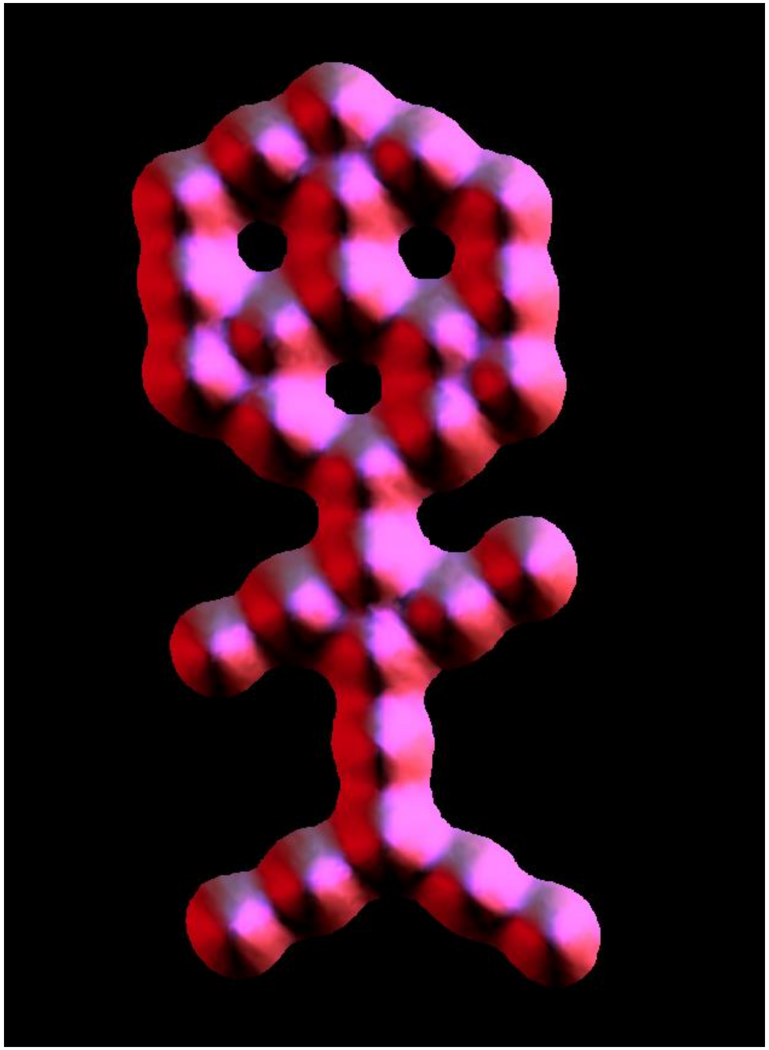


Study quantum chaos! (or at least try)

Fe on Cu(111)
4.5K UHV STM

M. Crommie et al, Science **262**, 218 (1993)

CO on Pt(111) 4.5K UHV STM



Quantum Mechanics

The spectacular success of Bohr's theory was followed by a series of disappointments.

Bohr's approach did not account for the emission spectra of atoms containing more than one electron, such as atoms of helium and lithium.

Nor did it explain why extra lines appear in the hydrogen emission spectrum when a magnetic field is applied.

Another problem arose with the discovery that electrons are wavelike: How can the "position" of a wave be specified?

We cannot define the precise location of a wave because a wave extends in space.

To describe the problem of trying to locate a subatomic particle that behaves like a wave $\Rightarrow$ Heisenberg uncertainty principle

Heisenberg uncertainty principle: it is impossible to know simultaneously both the momentum p (defined as mass times velocity) and the position of a particle with certainty.

Quantum Mechanics (1)

Heisenberg uncertainty principle:

it is impossible to know simultaneously both the momentum p (defined as mass times velocity) and the position of a particle with certainty.

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

Δx and Δp are the uncertainties in measuring the position and momentum of the particle, respectively.

If the measured uncertainties of position and momentum are large (say, in a crude experiment), their product can be substantially greater than $h/4\pi$ (hence the > sign).

Applying the Heisenberg uncertainty principle to the hydrogen atom, we see that in reality the electron does not orbit the nucleus in a well-defined path, as Bohr thought. If it did, we could determine precisely both the position of the electron (from its location on a particular orbit) and its momentum (from its kinetic energy) at the same time, a violation of the uncertainty principle.

Example 7.6

(a) An electron is moving at a speed of $8.0 \times 10^6\ \mathrm{m/s}$. If the uncertainty in measuring the speed is 1.0 percent of the speed, calculate the uncertainty in the electron's position. The mass of the electron is $9.1094 \times 10^{-31}\ \mathrm{kg}$.

(b) A baseball of mass 0.15 kg thrown at 100 mph has a momentum of $6.7\ \mathrm{kg \cdot m/s}$. If the uncertainty in measuring this momentum is 1.0×10^{-7} of the momentum, calculate the uncertainty in the baseball's position.

Solution

(a)The uncertainty in the electron's speed *u* is

$$\Delta u = 0.010 \times 8.0 \times 10^6\ \mathrm{m/s} = 8.0 \times 10^4\ \mathrm{m/s}$$

$$p = mu \Rightarrow \Delta p = m\Delta u = 9.1094 \times 10^{-31}\ \mathrm{kg} \times 8.0 \times 10^4\ \mathrm{m/s}$$

$$= 7.3 \times 10^{-26}\ \mathrm{kg \cdot m/s}$$

The uncertainty in the electron's position is

$$\Delta x = \frac{h}{4\pi\Delta p} = \frac{6.63 \times 10^{-34}\ \mathrm{J \cdot s}}{4\pi(7.3 \times 10^{-26}\ \mathrm{kg \cdot m/s})} = 7.2 \times 10^{-10}\ \mathrm{m}$$

This uncertainty corresponds to about 4 atomic diameters.

Example 7.6 (1)

(b) The uncertainty in the position of the baseball is

$$\Delta x = \frac{h}{4\pi \Delta p} = \frac{6.63 \times 10^{-34}\ \mathrm{J} \cdot s}{4\pi \times 1.0 \times 10^{-7} \times 6.7\ \mathrm{kg} \cdot \mathrm{m}/s()} = 7.9 \times 10^{-29}\ \mathrm{m}$$

This is such a small number as to be of no consequence, that is, there is practically no uncertainty in determining the position of the baseball in the macroscopic world.

Schrodinger Wave Equation

$$-\frac{\hbar^2}{2m}\nabla^2\Psi(\mathbf{r},t)+V(\mathbf{r},t)\Psi(\mathbf{r},t)=i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t)$$

$$\Psi(\mathbf{r},t)=\psi(\mathbf{r})\varphi(t) \qquad -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r})+V(\mathbf{r})\psi(\mathbf{r})=E\psi(\mathbf{r})$$

字母	名字						對應腓尼基字母
	古希臘語	現代希臘語	英語			中文	
			名稱	英式發音	美式發音		
Φ φ	φεῖ /pʰê:/	φι /ˈfi/	Phi	/ˈfaɪ/		斐	𐤒 (Qoph)
Ψ ψ	ψεῖ /psê:/	ψι /ˈpsi/	Psi	/ˈsaɪ/，/ˈpsaɪ/		普西	𐤒 (Qoph)

Schrodinger Wave Equation

In 1926 Schrodinger wrote an equation that described both the particle and wave nature of the e^-

$$-\frac{\hbar^2}{2m}\nabla^2\Psi(\mathbf{r},t)+V(\mathbf{r},t)\Psi(\mathbf{r},t)=i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t)$$

$$\because \quad \Psi(\mathbf{r},t)=\psi(\mathbf{r})\varphi(t) \;\Rightarrow\; -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r})+V(\mathbf{r})\psi(\mathbf{r})=E\psi(\mathbf{r})$$

The wave function Ψ itself has no direct physical meaning.

The probability of finding the electron in a certain region in space is proportional to the square of the wave function, Ψ^2.

Wave function (ψ) describes:

1. energy of e^- with a given ψ
2. probability of finding e^- in a volume of space

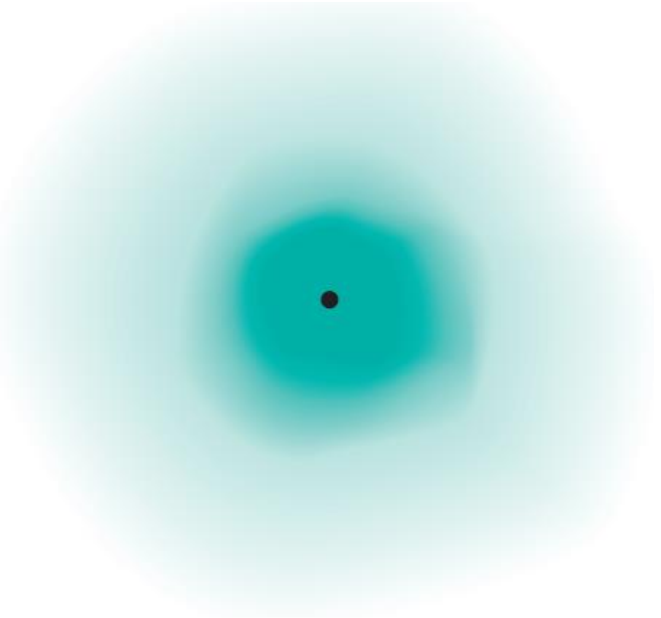
Schrodinger's equation can only be solved exactly for the hydrogen atom. Must approximate its solution for multi-electron systems.

Schrodinger Wave Equation (1)

The concept of electron density gives the probability that an electron will be found in a particular region of an atom.

The square of the wave function, Ψ^2, defines the distribution of electron density in three-dimensional space around the nucleus.

Regions of high electron density represent a high probability of locating the electron, whereas the opposite holds for regions of low electron density



A representation of the electron density distribution surrounding the nucleus in the hydrogen atom.

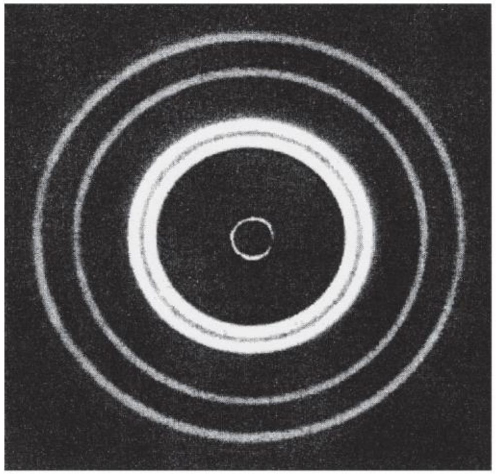
It shows a high probability of finding the electron closer to the nucleus.

Schrodinger Wave Equation (2)

The concept of electron density gives the probability that an electron will be found in a particular region of an atom.

The square of the wave function, Ψ^2, defines the distribution of electron density in three-dimensional space around the nucleus.

Regions of high electron density represent a high probability of locating the electron, whereas the opposite holds for regions of low electron density



©Education Development Center, Inc.

Schrodinger Wave Equation (3)

To distinguish the quantum mechanical description of an atom from Bohr's model, we speak of an atomic orbital, rather than an orbit. An atomic orbital can be thought of as the wave function (Ψ) of an electron in an atom.

When we say that an electron is in a certain orbital, we mean that the distribution of the electron density or the probability of locating the electron in space is described by the square of the wave function (Ψ^2) associated with that orbital.

An atomic orbital, therefore, has a characteristic energy, as well as a characteristic distribution of electron density.

The behavior of electrons in many-electron atoms (that is, atoms containing two or more electrons) is not the same as in the hydrogen atom. Assume that the difference is probably not too great. Thus, we can use the energies and wave functions obtained from the hydrogen atom as good approximations of the behavior of electrons in more complex atoms.

Schrodinger Wave Equation (4)

In quantum mechanics, three quantum numbers (n, ℓ, m_l) are required to describe the distribution of electrons in hydrogen and other atoms.

These numbers are derived from the mathematical solution of the Schrödinger equation for the hydrogen atom.

They are called the principal quantum number (n), the angular momentum quantum number (ℓ), and the magnetic quantum number (m_l). These quantum numbers label electrons will be used to describe atomic orbitals and to that reside in them.

A fourth quantum number—the spin quantum number (m_s)—describes the behavior of a specific electron and completes the description of electrons in atoms.

Schrodinger Wave Equation

Table 7B.1 The Schrödinger equation

Expression	Equation	Comment
Time-independent Schrödinger equation	$\hat{H}\psi = E\psi$	General case
	$-\dfrac{\hbar^2}{2m}\dfrac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x)$	One dimension
	$-\dfrac{\hbar^2}{2m}\left(\dfrac{\partial^2\psi}{\partial x^2} + \dfrac{\partial^2\psi}{\partial y^2}\right) + V(x,y)\psi(x,y) = E\psi(x,y)$	Two dimensions
	$\boxed{-\dfrac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi}$	Three dimensions
Laplacian operator	$\nabla^2 = \dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + \dfrac{\partial^2}{\partial z^2}$	
	$\boxed{\nabla^2 = \dfrac{1}{r}\dfrac{\partial^2}{\partial r^2}r + \dfrac{1}{r^2}\Lambda^2}$ $= \dfrac{\partial^2}{\partial r^2} + \dfrac{2}{r}\dfrac{\partial}{\partial r} + \dfrac{1}{r^2}\Lambda^2$ $= \dfrac{1}{r^2}\dfrac{\partial}{\partial r}r^2\dfrac{\partial}{\partial r} + \dfrac{1}{r^2}\Lambda^2$	Alternative forms
Legendrian operator	$\boxed{\Lambda^2 = \dfrac{1}{\sin^2\theta}\dfrac{\partial^2}{\partial\phi^2} + \dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\sin\theta\dfrac{\partial}{\partial\theta}}$	
Time-dependent Schrödinger equation	$\hat{H}\Psi = i\hbar\dfrac{\partial\Psi}{\partial t}$	

$$-\frac{\hbar^2}{2m}\nabla^2\Psi(\mathbf{r},t) + V(\mathbf{r},t)\Psi(\mathbf{r},t) = i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t)$$

$$\Psi(\mathbf{r},t) = \psi(\mathbf{r})\varphi(t)$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

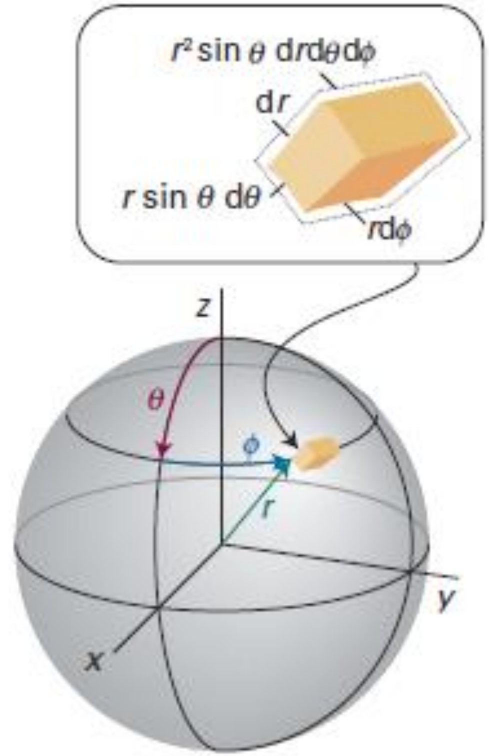


Fig. 7.21 The spherical polar coordinates used for discussing systems with spherical symmetry.

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \xrightarrow{A=B} \psi(x) = A\left(e^{ikx} + e^{-ikx}\right) = 2A\cos kx$$

$$\Rightarrow \psi = \underbrace{\psi_{\rightarrow}}_{\substack{\text{Particle with linear} \\ \text{momentum } +k\hbar}} + \underbrace{\psi_{\leftarrow}}_{\substack{\text{Particle with linear} \\ \text{momentum } -k\hbar}}$$

Suppose the wavefunction is known to be a superposition of many different linear momentum eigenfunctions and written as: the linear combination :

$$\boxed{\psi = c_1\psi_1 + c_2\psi_2 + \cdots = \sum_k c_k\psi_k \quad \text{where} \begin{cases} c_k \equiv \text{numerical (possibly complex) coefficients} \\ \psi_k \equiv \text{correspond to different momentum states.} \end{cases}}$$

The functions ψ_k are said to form a <u>complete set</u> in the sense that any arbitrary function can be expressed as a linear combination of them.

quantum mechanics:

1. When the momentum is measured, in a single observation one of the eigenvalues corresponding to the ψ_k that contribute to the superposition will be found.
2. The probability of measuring a particular eigenvalue in a series of observations is proportional to the square modulus ($|c_k|^2$) of the corresponding coefficient in the linear combination.
3. The average value of a large number of observations is given by the expectation value, $<\Omega>$, of the operator corresponding to the observable of interest.

The expectation value of an operator $\hat{\Omega}$ is defined as $<\Omega> = \int \psi^* \hat{\Omega}\psi \mathrm{d}\tau$

(valid only for *normalized wavefunctions*)

An expectation value is the weighted average of a large number of observations of a property.

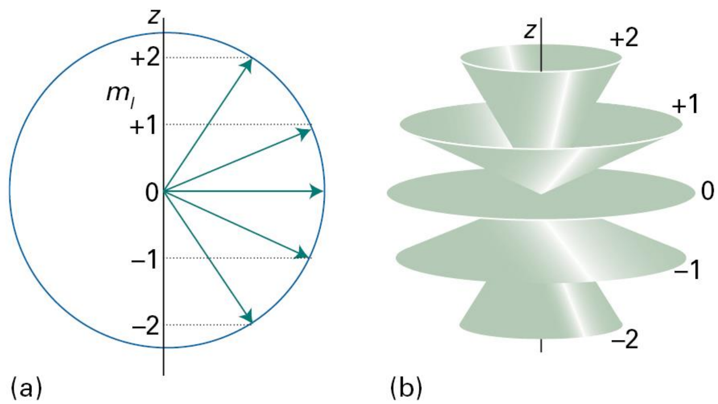


Figure 8C.11 (a) A summary of Fig. 8C.9. However, because the azimuthal angle of the vector around the *z*-axis is indeterminate, a better representation is as in (b), where each vector lies at an unspecified azimuthal angle on its cone.

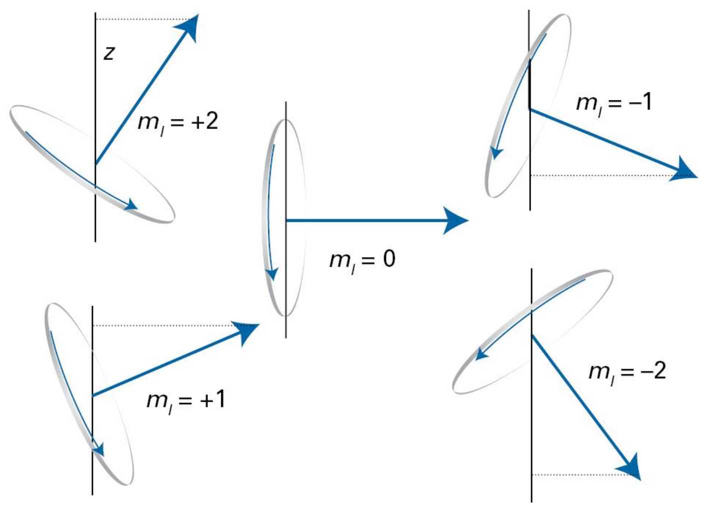


Figure 8C.9 The permitted orientations of angular momentum when $l = 2$. We shall see soon that this representation is too specific because the azimuthal orientation of the vector (its angle around *z*) is indeterminate.

Table 8C.1 The spherical harmonics, $Y_{l,m_l}(\theta,\phi)$

l		m_l	$Y_{l,m_l}(\theta,\phi)$
0	*s*	0	$\left(\frac{1}{4\pi}\right)^{1/2}$
1	*p*	0	$\left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$
		±1	$\mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta\,e^{\pm i\phi}$
2	*d*	0	$\left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta-1)$
		±1	$\mp\left(\frac{15}{8\pi}\right)^{1/2}\cos\theta\sin\theta\,e^{\pm i\phi}$
		±2	$\left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta\,e^{\pm 2i\phi}$
3	*f*	0	$\left(\frac{7}{16\pi}\right)^{1/2}(5\cos^3\theta-3\cos\theta)$
		±1	$\mp\left(\frac{21}{64\pi}\right)^{1/2}(5\cos^2\theta-1)\sin\theta\,e^{\pm i\phi}$
		±2	$\left(\frac{105}{32\pi}\right)^{1/2}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$
		±3	$\mp\left(\frac{35}{64\pi}\right)^{1/2}\sin^3\theta\,e^{\pm 3i\phi}$

Schrodinger Wave Equation (5)

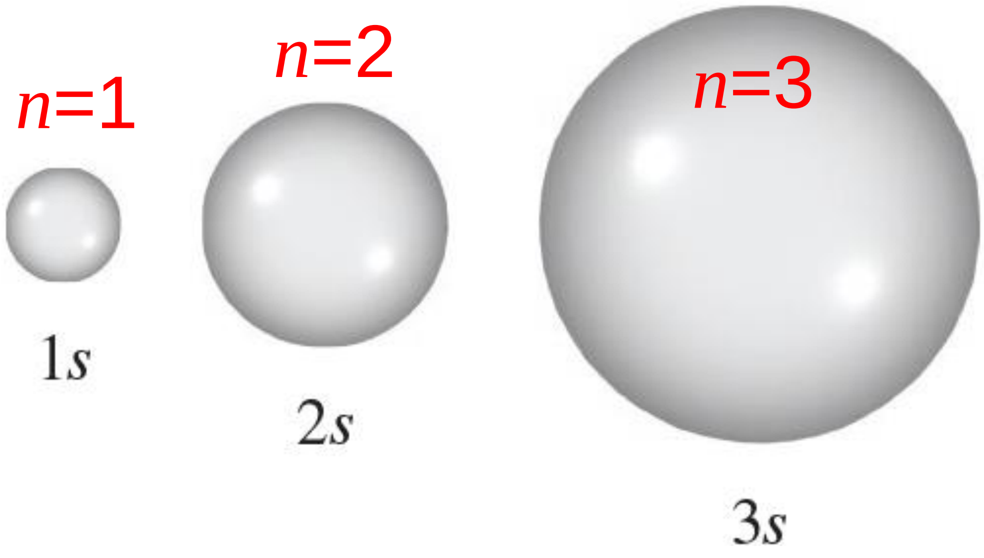
ψ is a function of four numbers called ***quantum numbers*** (n, ℓ, m_ℓ, m_s)

principal quantum number n

$n = 1, 2, 3, 4, \ldots.$

$$E_n = -R_H\left(\frac{1}{n^2}\right)$$

distance of e^- from the nucleus



Schrodinger Wave Equation (6)

quantum numbers: (n, ℓ, m_ℓ, m_s)

angular momentum quantum number ℓ

for a given value of n, $\ell = 0, 1, 2, 3, \ldots\ n - 1$

$n = 1$, $\ell = 0$

$n = 2$, $\ell = 0$ **or** 1

$n = 3$, $\ell = 0, 1,$ **or** 2

$\ell = 0$ *s* orbital

$\ell = 1$ *p* orbital

$\ell = 2$ *d* orbital

$\ell = 3$ *f* orbital

ℓ	0	1	2	3	4	5
Name of orbital	*s*	*p*	*d*	*f*	*g*	*h*

$\ell \Rightarrow$the Shape of the orbitals–the "volume" of space that the e^- occupies

Schrodinger Wave Equation (7)

quantum numbers: (n, ℓ, m_ℓ, m_s)

A collection of orbitals with the same value of n $\Rightarrow$shell.

One or more orbitals with the same n and ℓ values $\Rightarrow$subshell.

<example> the shell with n=2 is composed of two subshells, ℓ = 0 and 1 (the allowed values for n=2).

These subshells are called the 2s and 2p subshells where 2 denotes the value of n, and s and p denote the values of ℓ.

Schrodinger Wave Equation (8)

quantum numbers: (n, ℓ, m_ℓ, m_s)

magnetic quantum number m_ℓ

for a given value of ℓ

$$m_\ell = \underbrace{-\ell, \ldots\ldots, 0, \ldots\ldots +\ell}_{2\ell+1}$$

if $\ell = 0$ (s orbital), $m_\ell = 0$:2×0+1=1

if $\ell = 1$ (p orbital), $m_\ell = -1, 0,$ **or** 1 :2×1+1=3

if $\ell = 2$ (d orbital), $m_\ell = -2, -1, 0, 1,$ **or** 2 :2×2+1=5

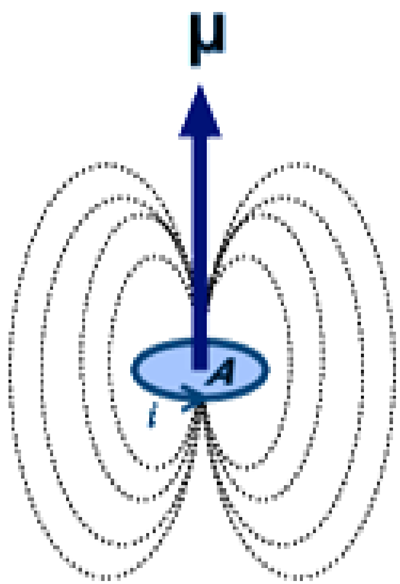
$\boxed{m_\ell \Rightarrow \text{The orientation of the orbital in space}}$

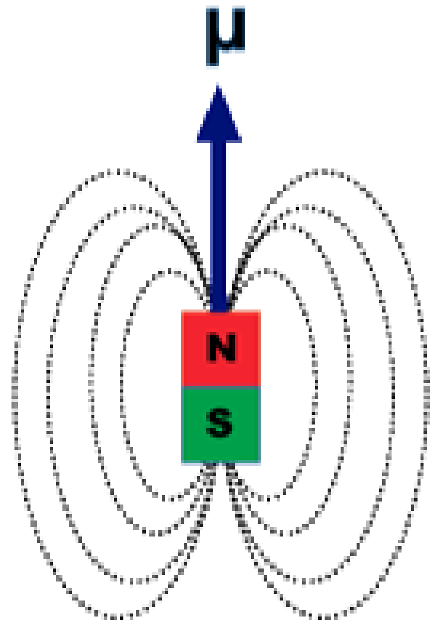
<example> n=2 and ℓ=1.

We have a 2p subshell, and in this subshell we have three 2p orbitals (there are three values of m_ℓ= –1, 0, and 1).

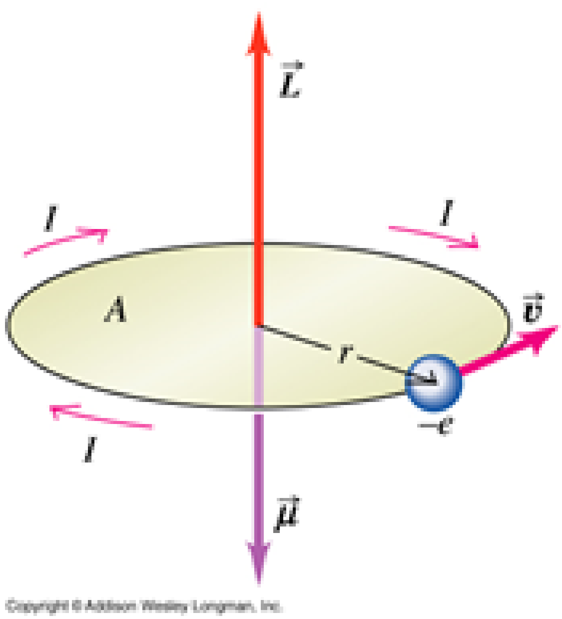
Magnetic dipoles

Macroscale





Atomic scale

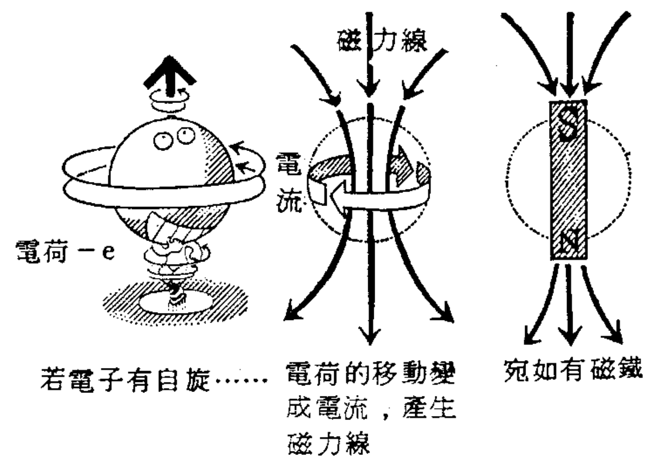


磁性的來源與基本現象

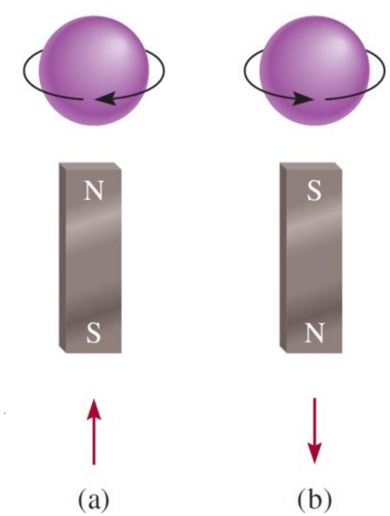
磁性本源 ⟹ 量子力學 狹義相對論 ⟹ 電子的自旋(spin)

電子的自旋磁矩 (magnetic moment) $\vec{\mu}_B = \dfrac{eh}{4\pi\, m_e c}$.

Quantum ***Classical***

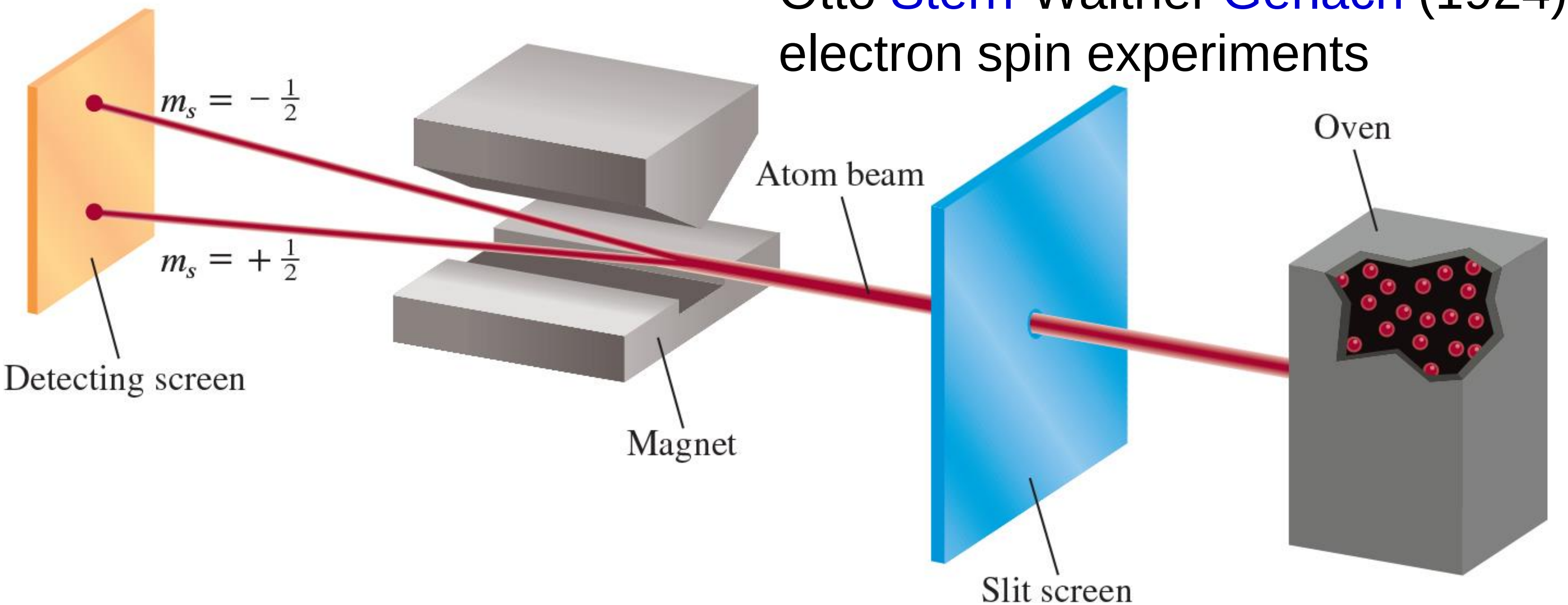


若電子轉動，即形成磁場。



Schrodinger Wave Equation (9)

Otto Stern-Walther Gerlach (1924) electron spin experiments



Experimental arrangement for demonstrating the spinning motion of electrons. A beam of atoms is directed through a magnetic field. For example, when a hydrogen atom with a single electron passes through the field, it is deflected in one direction or the other, depending on the direction of the spin.

In a stream consisting of many atoms, there will be equal distributions of the two kinds of spins, so that two spots of equal intensity are detected on the screen.

Schrodinger Wave Equation (9)

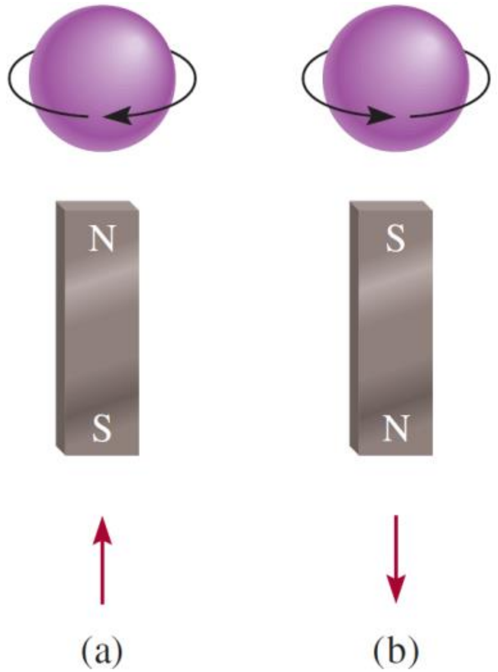
(n, l, m_ℓ, m_s)

spin quantum number m_s

$m_s = +{}^1/_2$ or $-{}^1/_2$

According to electromagnetic theory, a spinning charge generates a magnetic field, and it is this motion that causes an electron to behave like a magnet.

A beam of gaseous atoms generated in a hot furnace passes through a nonhomogeneous magnetic field. The interaction between an electron and the magnetic field causes the atom to be deflected from its straight-line path. Because the spinning motion is completely random, the electrons in half of the atoms will be spinning in one direction, and those atoms will be deflected in one way; the electrons in the other half of the atoms will be spinning in the opposite direction, and those atoms will be deflected in the other direction. Thus, two spots of equal intensity are observed on the detecting screen.



$m_s = +{}^1/_2$ $m_s = -{}^1/_2$

Quantum Numbers and Atomic Orbitals

if $\ell = 0, m_\ell = 0$: one s orbital

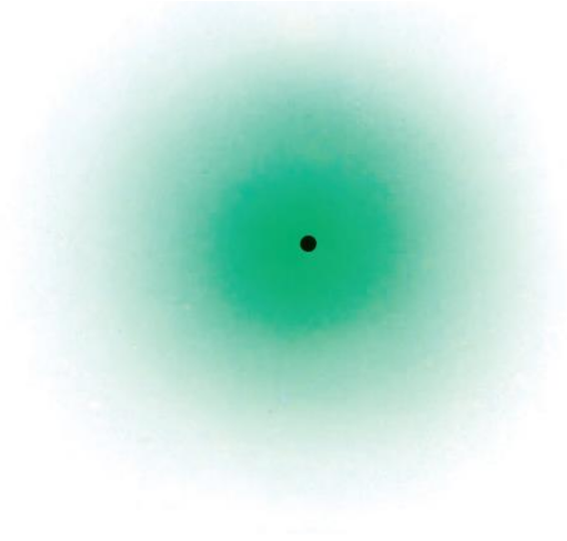
if $\ell = 1, m_\ell = -1, 0, 1$: three p orbitals–p_x, p_y, p_z

if $\ell = 2, m_\ell = -2, -1, 0, 1, 2$: five d orbitals–d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$, d_{z^2}

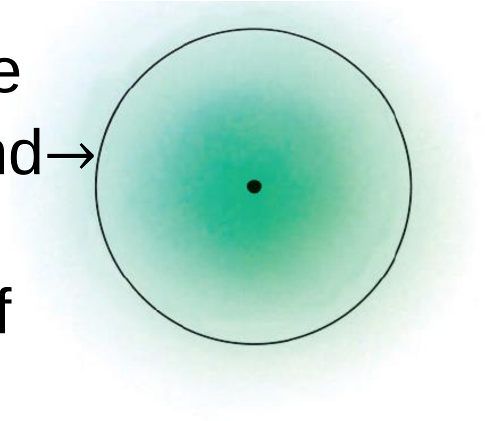
Table 7.2 Relation Between Quantum and Atomic Orbitals

n	ℓ	m_ℓ	Number of Orbitals	Atomic Orbital Designations
1	0	0	1	$1s$
2	0	0	1	$2s$
	1	$-1, 0, 1$	3	$2p_x, 2p_y, 2p_z$
3	0	0	1	$3s$
	1	$-1, 0, 1$	3	$3p_x, 3p_y, 3p_z$
	2	$-2, -1, 0, 1, 2$	5	$3d_{xy}, 3d_{yz}, 3d_{xz},$ $3d_{x^2-y^2}, 3d_{z^2}$
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

Probability Density



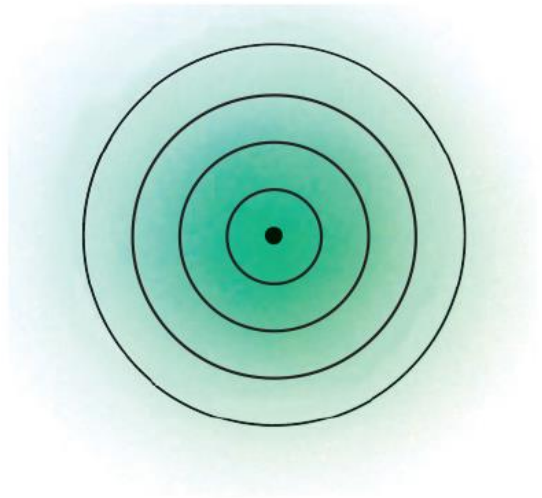
Where 90% of the e^- density is found→ for the 1s orbital within a sphere of radius 100 pm surrounding the nucleus.

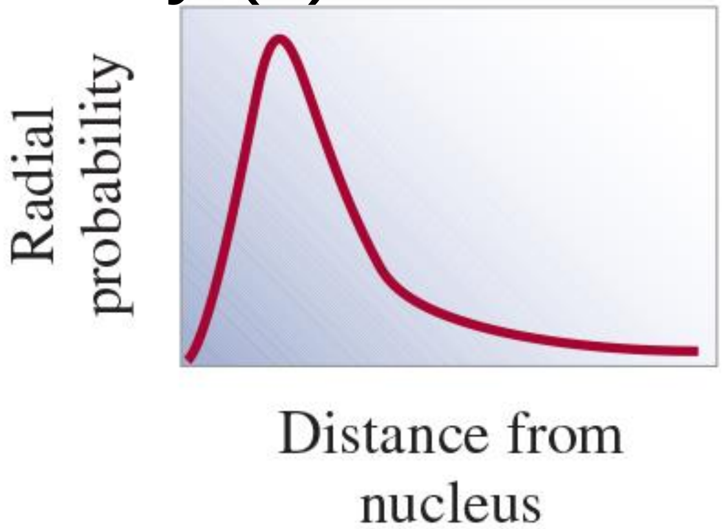


(a) Plot of electron density in the hydrogen 1s orbital as a function of the distance from the nucleus. The electron density falls off rapidly as the distance from the nucleus increases.

(b) Boundary surface diagram of the hydrogen 1s orbital.

Probability Density (1)





(c) A more realistic way of viewing electron density distribution is to divide the 1s orbital into successive spherical thin shells. A plot of the probability of finding the electron in each shell, called radial probability, as a function of distance shows a maximum at 52.9 pm from the nucleus. Interestingly, this is equal to the radius of the innermost orbit in the Bohr model.

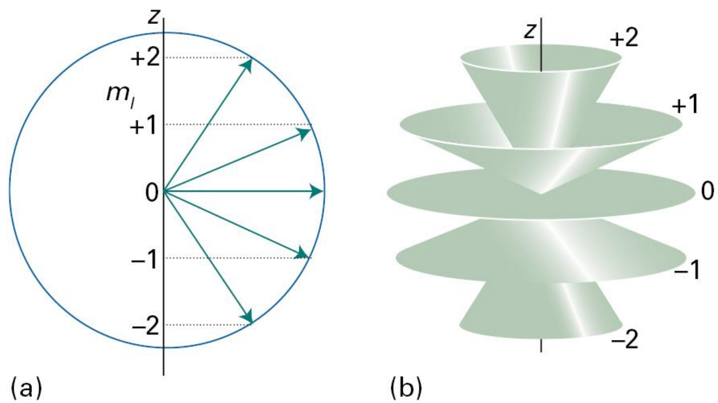


Figure 8C.11 (a) A summary of Fig. 8C.9. However, because the azimuthal angle of the vector around the *z*-axis is indeterminate, a better representation is as in (b), where each vector lies at an unspecified azimuthal angle on its cone.

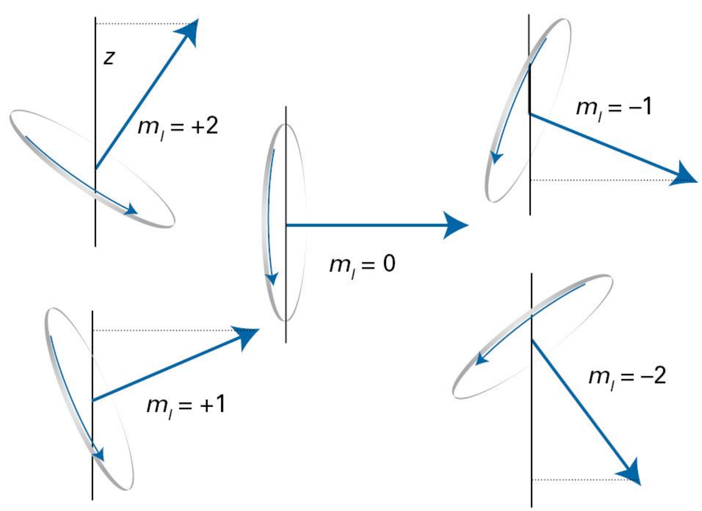


Figure 8C.9 The permitted orientations of angular momentum when $l=2$. We shall see soon that this representation is too specific because the azimuthal orientation of the vector (its angle around *z*) is indeterminate.

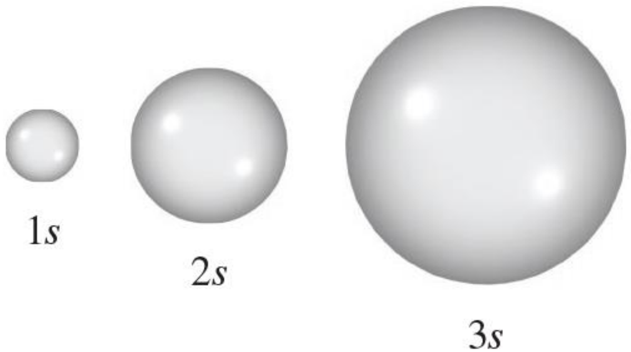
Table 8C.1 The spherical harmonics, $Y_{l,m_l}(\theta,\phi)$

l		m_l	$Y_{l,m_l}(\theta,\phi)$
0	*s*	0	$\left(\dfrac{1}{4\pi}\right)^{1/2}$
1	*p*	0	$\left(\dfrac{3}{4\pi}\right)^{1/2}\cos\theta$
		±1	$\mp\left(\dfrac{3}{8\pi}\right)^{1/2}\sin\theta\,e^{\pm i\phi}$
2	*d*	0	$\left(\dfrac{5}{16\pi}\right)^{1/2}(3\cos^2\theta-1)$
		±1	$\mp\left(\dfrac{15}{8\pi}\right)^{1/2}\cos\theta\sin\theta\,e^{\pm i\phi}$
		±2	$\left(\dfrac{15}{32\pi}\right)^{1/2}\sin^2\theta\,e^{\pm 2i\phi}$
3	*f*	0	$\left(\dfrac{7}{16\pi}\right)^{1/2}(5\cos^3\theta-3\cos\theta)$
		±1	$\mp\left(\dfrac{21}{64\pi}\right)^{1/2}(5\cos^2\theta-1)\sin\theta\,e^{\pm i\phi}$
		±2	$\left(\dfrac{105}{32\pi}\right)^{1/2}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$
		±3	$\mp\left(\dfrac{35}{64\pi}\right)^{1/2}\sin^3\theta\,e^{\pm 3i\phi}$

Atomic Orbitals

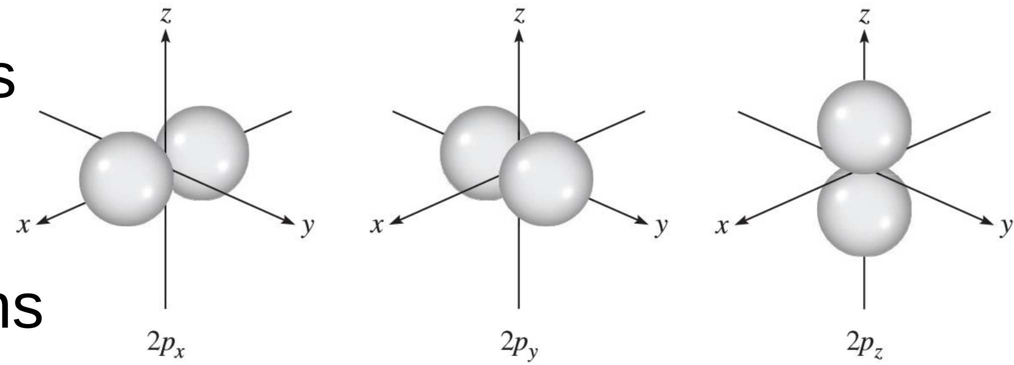
$l = 0$ (s orbitals)

Boundary surface diagrams of the hydrogen 1s, 2s, and 3s orbitals. Each sphere contains about 90 % of the total electron density. All s orbitals are spherical. Roughly speaking, the size of an orbital is proportional to n^2, where n is the principal quantum number.



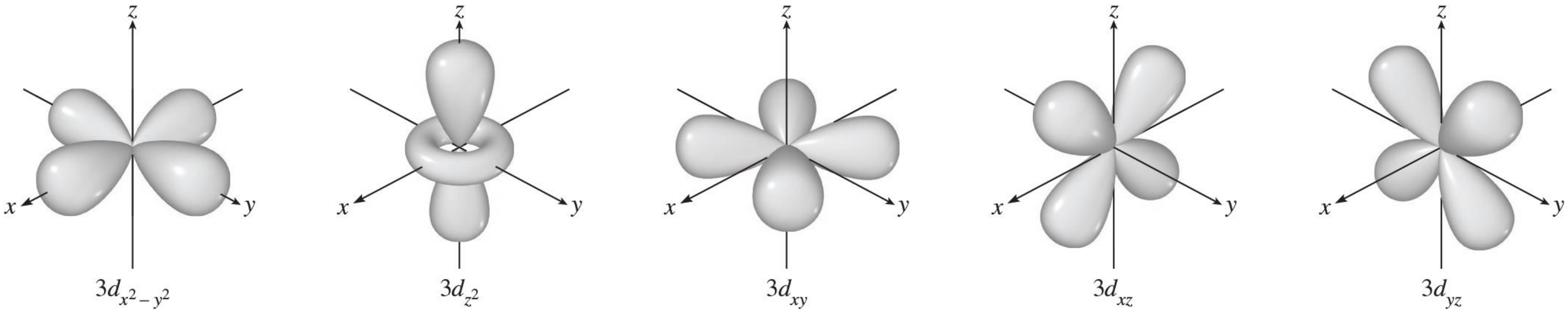
$l = 1$ (p orbitals)

The boundary surface diagrams of the three 2p orbitals. These orbitals are identical in shape and energy, but their orientations are different. The p orbitals of higher principal quantum numbers have similar shapes.



Atomic Orbitals (1)

$l = 2$ (d orbitals)



Boundary surface diagrams of the five 3*d* orbitals. Although the $3d_{z^2}$ orbital looks different, it is equivalent to the other four orbitals in all other respects. The *d* orbitals of higher principal quantum numbers have similar shapes.

Example 7.7

What are the allowed values of m_ℓ when n = 4 and ℓ = 1?

Solution The values of m_ℓ depend only on the value of ℓ, not on the value of n. The values of m_ℓ can vary from $-\ell$ to ℓ. Therefore, m_ℓ can be $-1, 0,$ or 1.

Example 7.8

List the values of n, ℓ, and m_ℓ for orbitals in the 4d subshell.

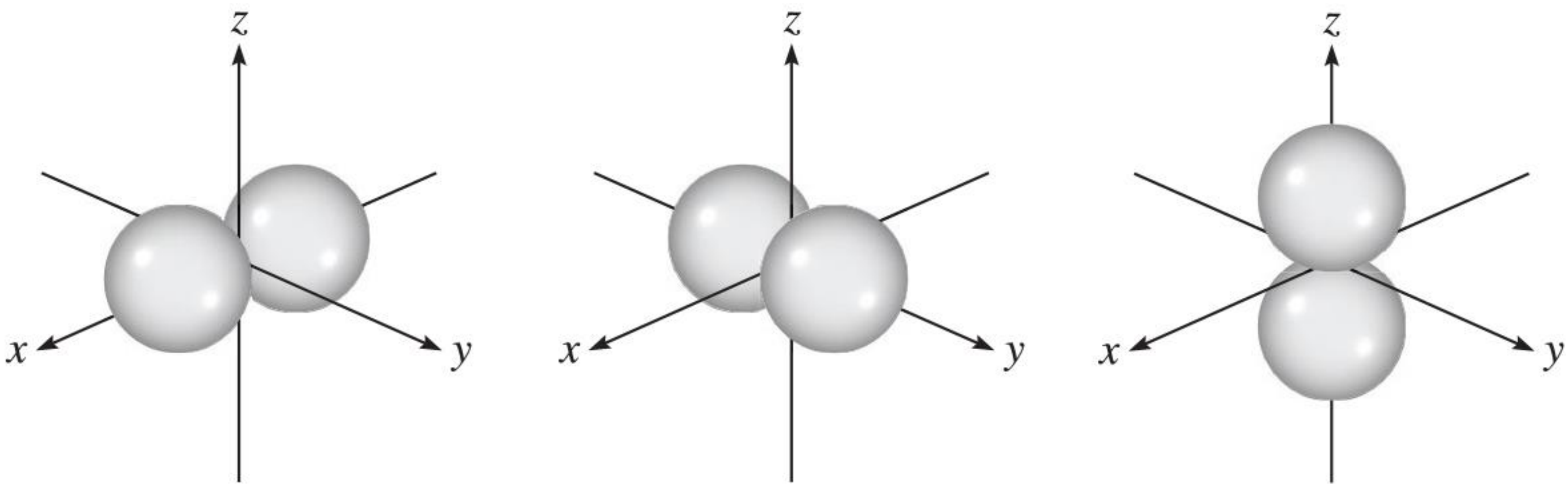
Solution

As we saw earlier, the number given in the designation of the subshell is the principal quantum number, so in this case $n = 4$. The letter designates the type of orbital. Because we are dealing with d orbitals, $\ell = 2$. The values of m_ℓ can vary from $-\ell$ to ℓ. Therefore, m_ℓ can be $-2, -1, 0, 1,$ or 2.

Orientation of Orbitals

$m_l = -1, 0,$ **or** 1

3 orientations in space

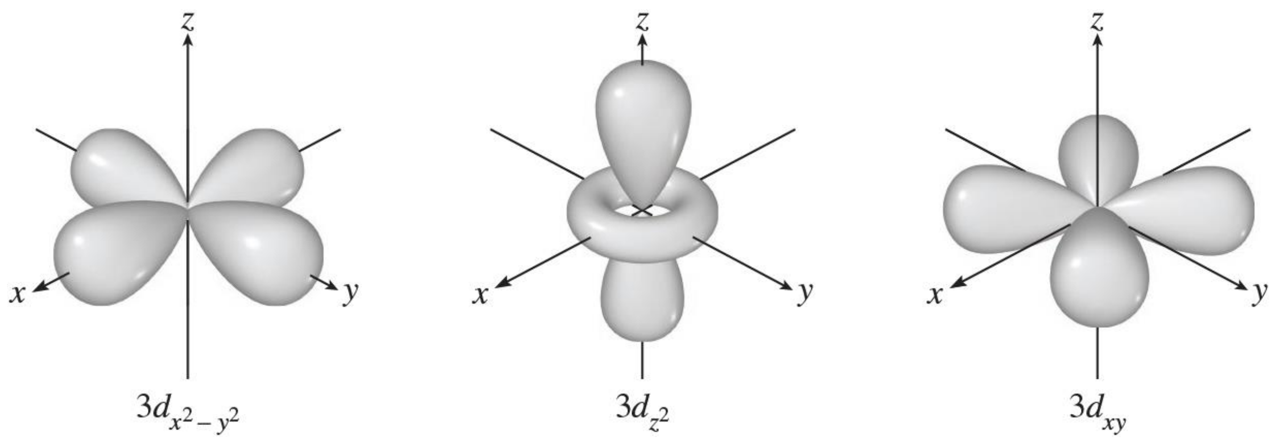


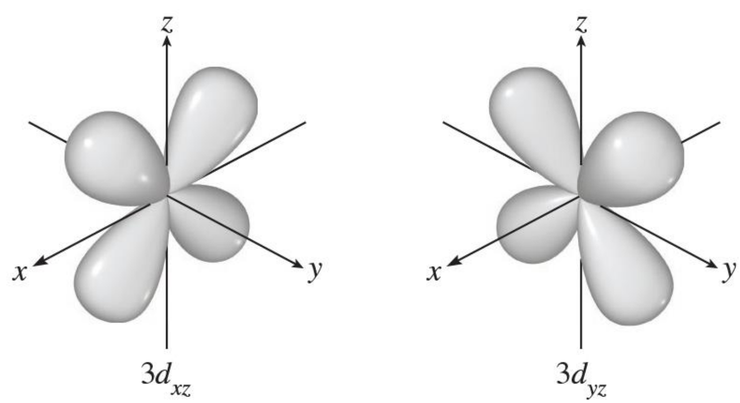
$2p_x$ $2p_y$ $2p_z$

Orientation of Orbitals (1)

$m_l = -2, -1, 0, 1,$ **or** 2

5 orientations in space





Example 7.9

What is the total number of orbitals associated with the principal quantum number $n = 3$?

Solution

For $n = 3$, the possible values of ℓ are 0, 1, and 2.

one 3*s* orbital $(n = 3, \ell = 0, \text{and } m_\ell = 0)$;

three 3*p* orbitals $(n = 3, \ell = 1, \text{and } m_\ell = -1, 0, 1)$;

five 3*d* orbitals $(n = 3, \ell = 2, \text{and } m_\ell = -2, -1, 0, 1, 2)$.

$\Rightarrow$The total number of orbitals is $1 + 3 + 5 = 9$.

Schrodinger Wave Equation (11)

quantum numbers: (n, ℓ, m_ℓ, m_s)

$\Rightarrow$label completely an electron in any orbital in any atom.

$\Rightarrow$specify the "address" of an electron in an atom.

Example: Find the four quantum numbers for a 2s orbital electron.

$\because n = 2, \ell = 0, m_\ell = 0, m_s = +\frac{1}{2} \text{ or } -\frac{1}{2}$

$\Rightarrow$The quantum numbers are either $(2, 0, 0, +\frac{1}{2})$ or $(2, 0, 0, -\frac{1}{2})$.

The value of m_s has no effect on the energy (n), size(n), shape(ℓ), or orientation of an orbital(m_ℓ), but it determines how electrons are arranged in an orbital.

Shell – electrons with the same value of n

Subshell – electrons with the same values of n **and** ℓ

Orbital – electrons with the same values of n, ℓ, **and** m_ℓ

Example 7.9

Write the four quantum numbers for an electron in a 3*p* orbital.

Solution

the principal quantum number n = 3 and the angular momentum quantum number ℓ =1 (because we are dealing with a *p* orbital).

For $\ell = 1$, there are three values of m_ℓ given by -1, 0, and 1.

Because the electron spin quantum number m_s can be either $+{}^1/_2$ or $-{}^1/_2$, we conclude that there are six possible ways to designate the electron using the (n, ℓ, m_ℓ, m_s) notation.

These are:

$$(3, 1, -1, +\tfrac{1}{2}) \qquad (3, 1, -1, -\tfrac{1}{2})$$
$$(3, 1, 0, +\tfrac{1}{2}) \qquad (3, 1, 0, -\tfrac{1}{2})$$
$$(3, 1, 1, +\tfrac{1}{2}) \qquad (3, 1, 1, -\tfrac{1}{2})$$

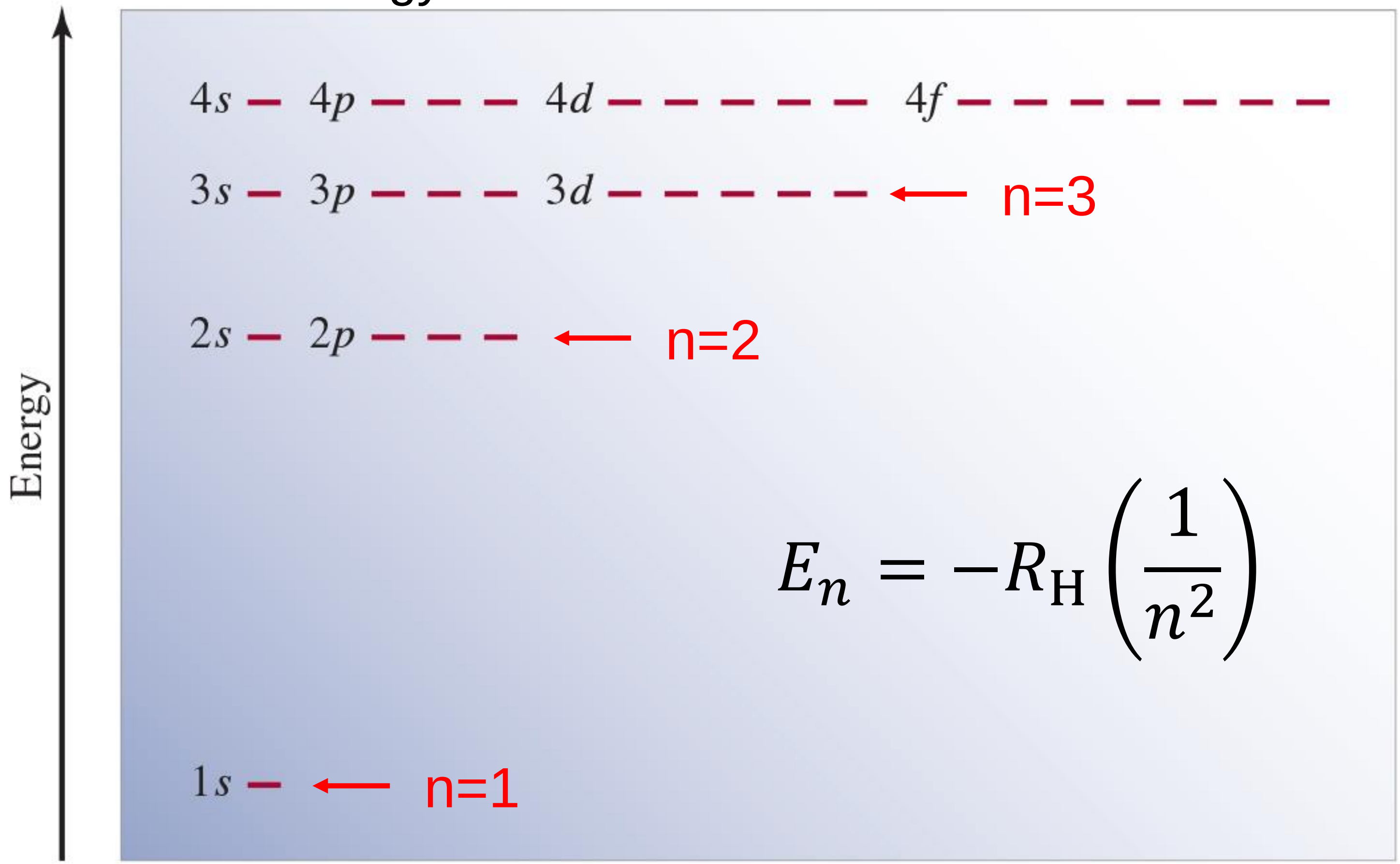
Energy of orbitals in a ***single*** electron atom

Energy only depends on principal quantum number ***n***

The energies of hydrogen orbitals increase as follows

$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < \cdots$$

Orbital energy levels in the **hydrogen atom**. Each short horizontal line represents one orbital. Orbitals with the same principal quantum number (n) all have the same energy.

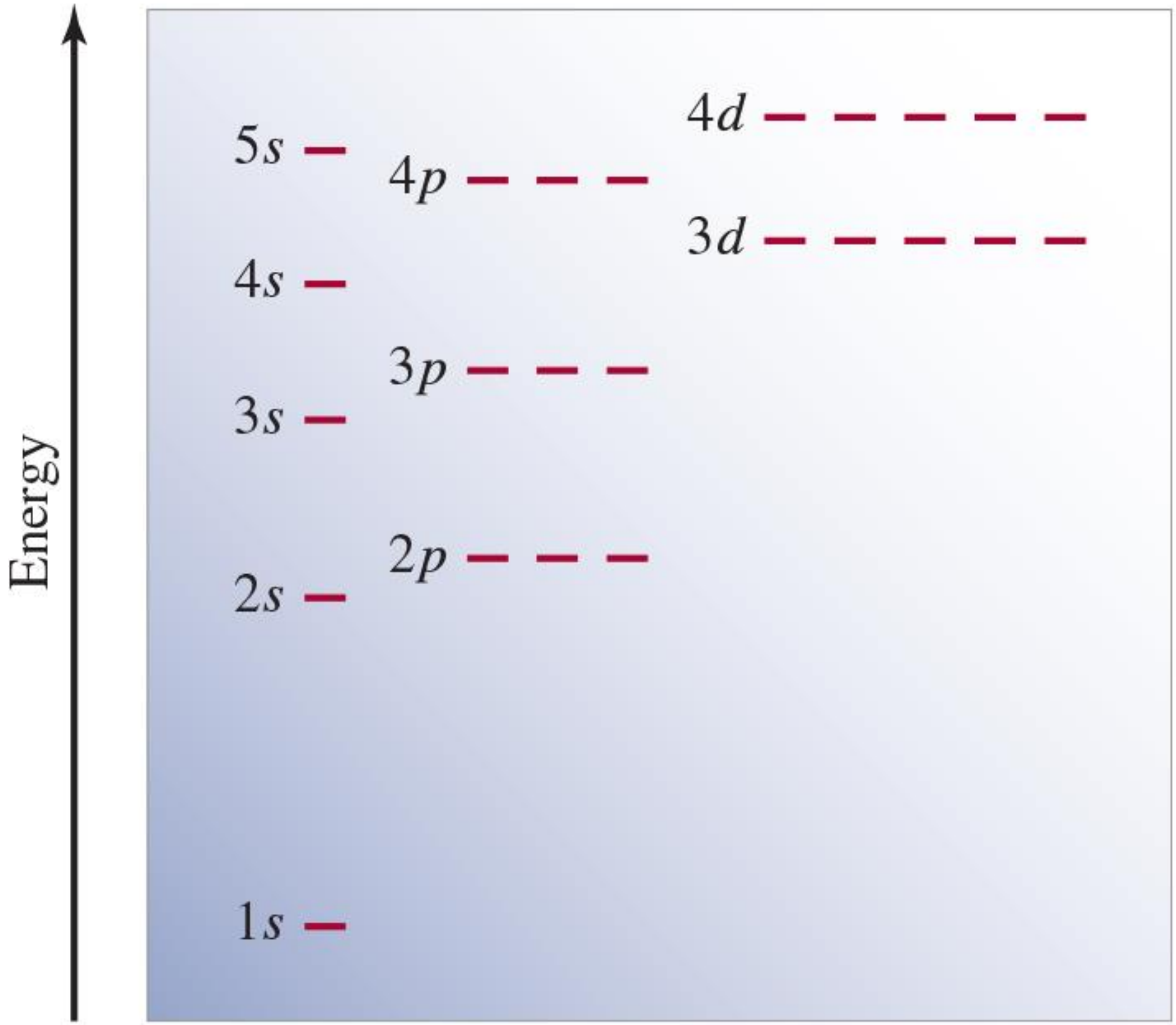


$$E_n = -R_{\mathrm{H}}\left(\frac{1}{n^2}\right)$$

Energy of orbitals in a ***multi***-electron atom

Energy depends on *n* and *ℓ*

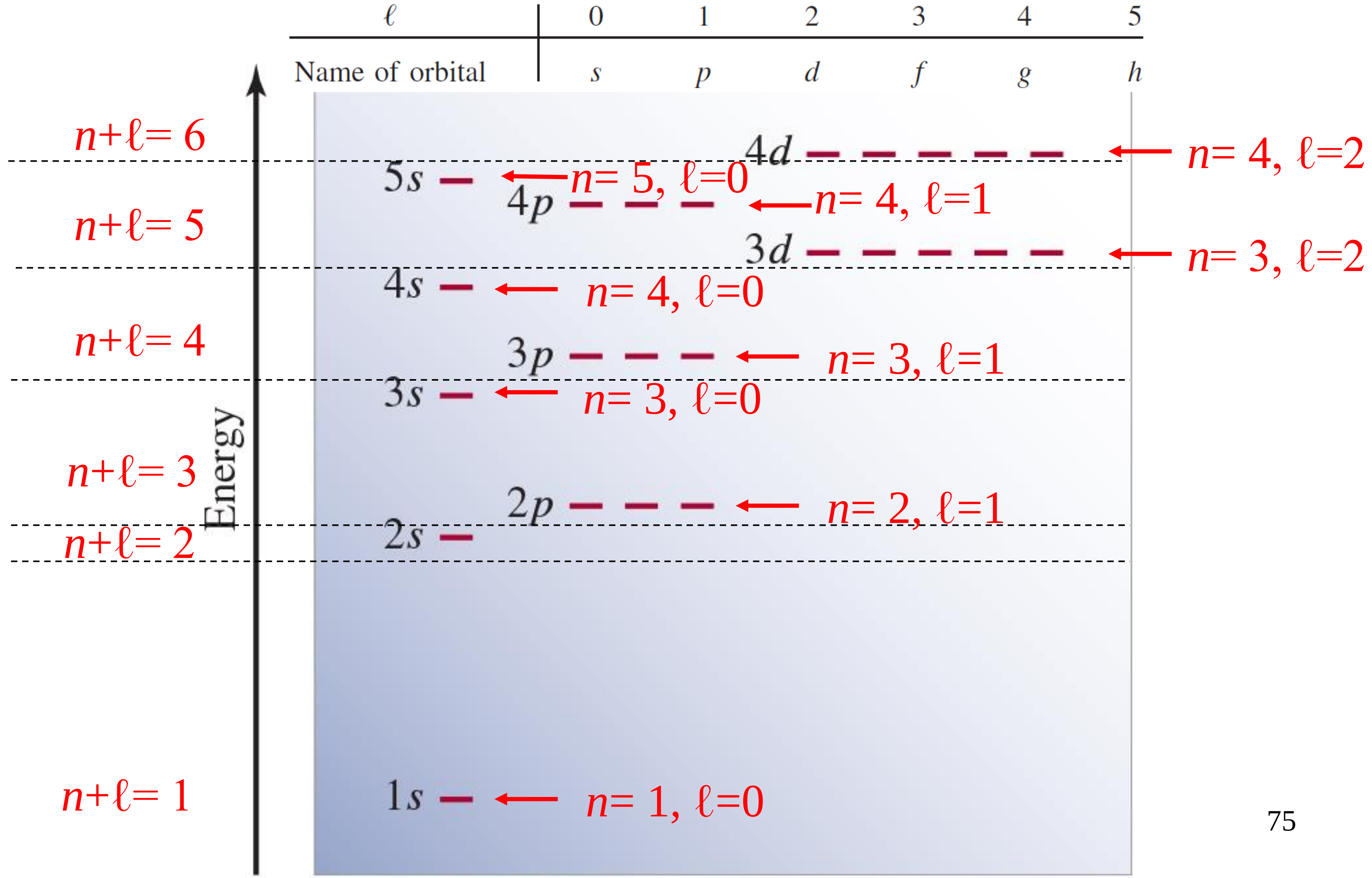
Orbital energy levels in a **many-electron atom**.
The energy level depends on both *n* and ℓ values.



Energy of orbitals in a ***multi***-electron atom

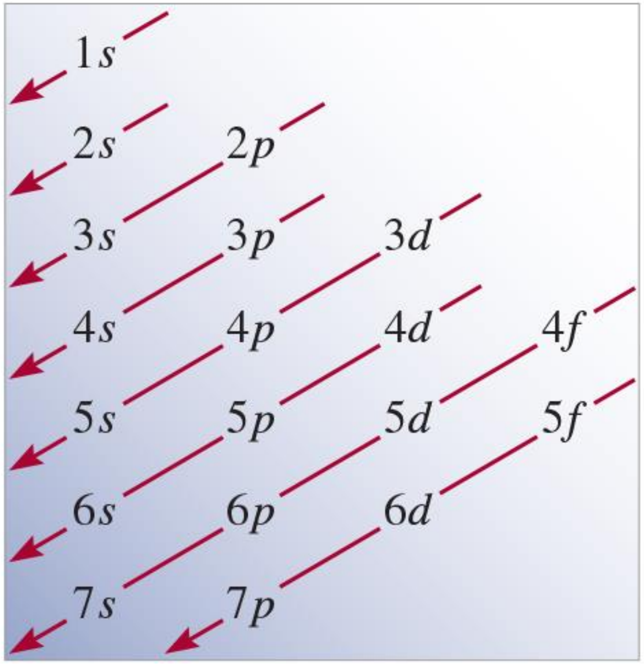
Energy depends on *n* and *ℓ*

Orbital energy levels in a **many-electron atom**.
The energy level depends on both *n* and ℓ values.



Order of orbitals (filling) in multi-electron atom

1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s



Electron Configuration

Electron configuration is how the electrons are distributed among the various atomic orbitals in an atom.

number of electrons in the orbital or subshell

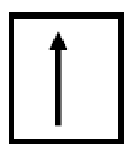
$1s^1$

principal quantum number n

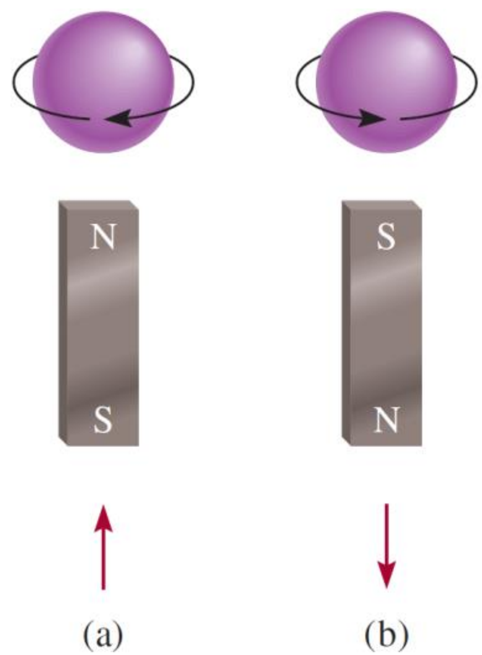
angular momentum quantum number ℓ

Orbital diagram

H



$1s^1$



Schrodinger Wave Equation (10)

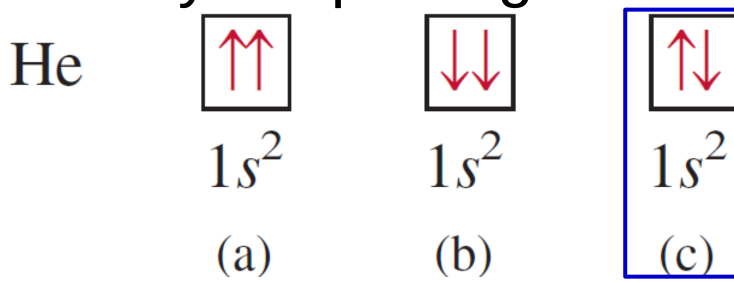
Pauli exclusion principle - no two electrons in an atom can have the same four quantum numbers:(n, ℓ, m_ℓ, m_s).

If two electrons in an atom should have the same $n, \ell,$ and m_ℓ values (these two electrons are in the same atomic orbital), then they must have different values of m_s.

$\Rightarrow$only two electrons may occupy the same atomic orbital, and these electrons must have opposite spins.

<Example> Helium atom

The three possible ways of placing two electrons in the 1s orbital are as follows:

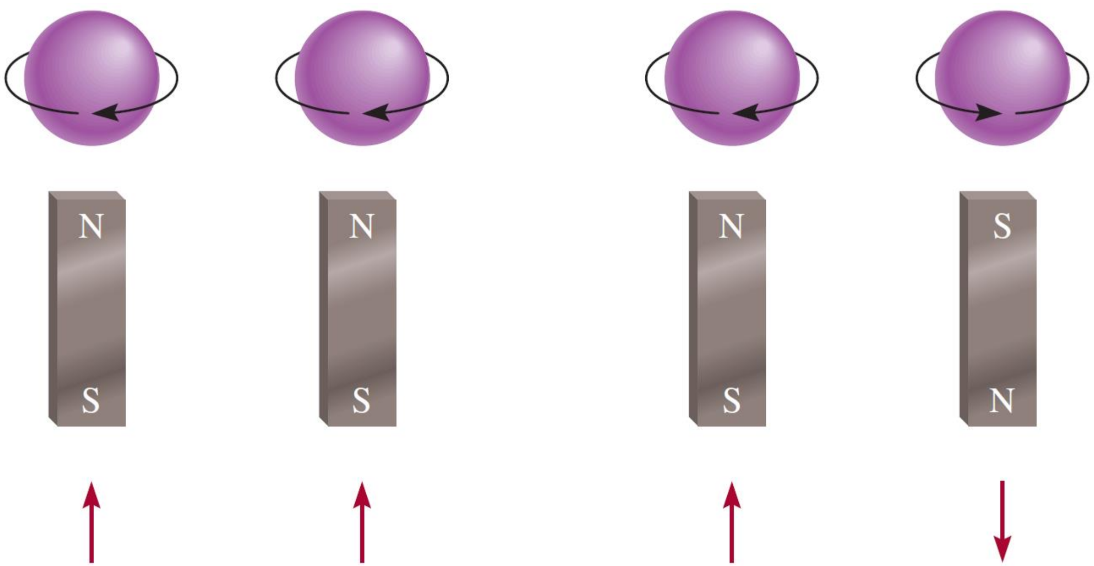


physically acceptable

$(1, 0, 0, +\frac{1}{2})$; $(1, 0, 0, -\frac{1}{2})$

Paramagnetism and Diamagnetism

The (a) parallel (↑↑ or ↓↓)and (b) antiparallel spins of two electrons.
In (a) the two magnetic fields reinforce each other.
In (b) the two magnetic fields cancel each other.



(a) (b)

Paramagnetic substances are those that contain net unpaired spins and are attracted by a magnet.

If the electron spins are paired, or antiparallel to each other (↑↓ or ↓↑), the magnetic effects cancel out. Diamagnetic substances do not contain net unpaired spins and are slightly repelled by a magnet.

Paramagnetic

unpaired electrons

↑ ↑ __

2p

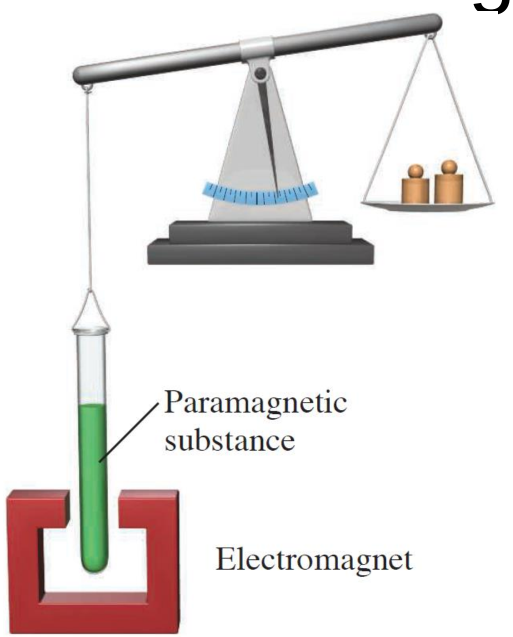
Diamagnetic

all electrons paired

↑↓ ↑↓ ↑↓

2p

Paramagnetism and Diamagnetism (1)



Initially the paramagnetic substance was weighed on a balance in the absence of a magnetic field.
When the electromagnet is turned on, the balance is offset because the sample tube is drawn into the magnetic field.

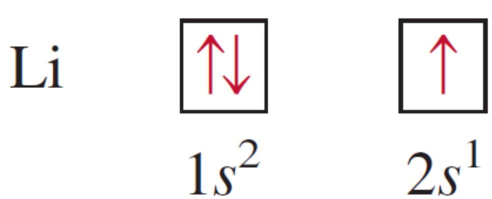
Knowing the concentration and the additional mass needed to reestablish balance, it is possible to calculate the number of unpaired electrons in the sample.

Any atom with an odd number of electrons will always contain one or more unpaired spins. On the other hand, atoms containing an even number of electrons may or may not contain unpaired spins.

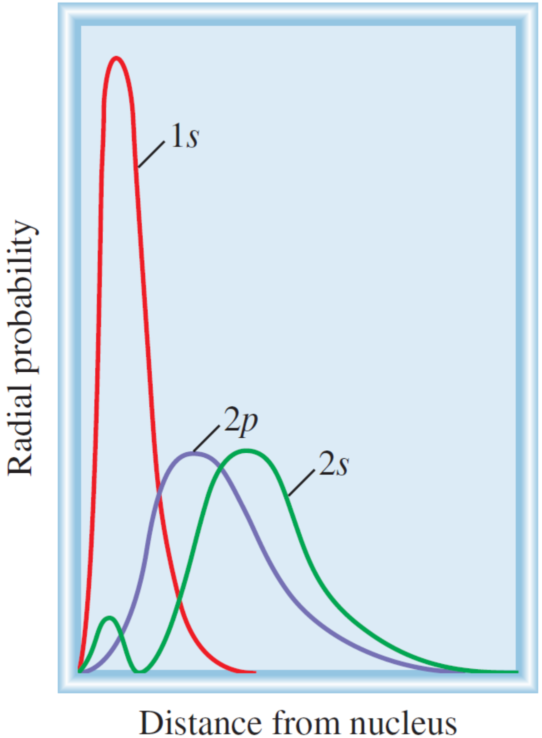
<Example> Lithium atom (Z=3)

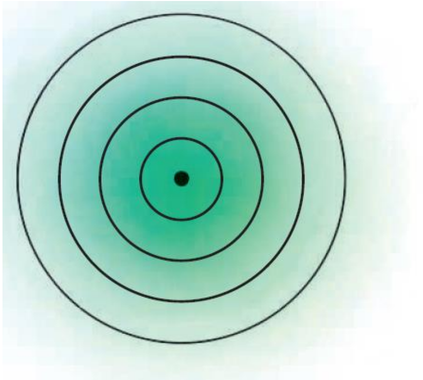
electron configuration: $1s^2 2s^1$

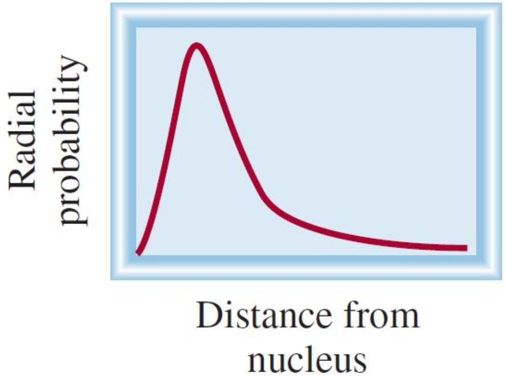
orbital diagram:



The Shielding Effect in Many-Electron Atoms

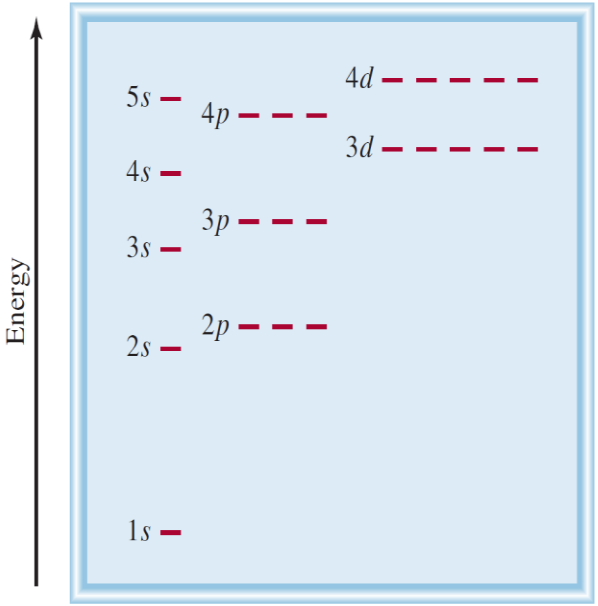






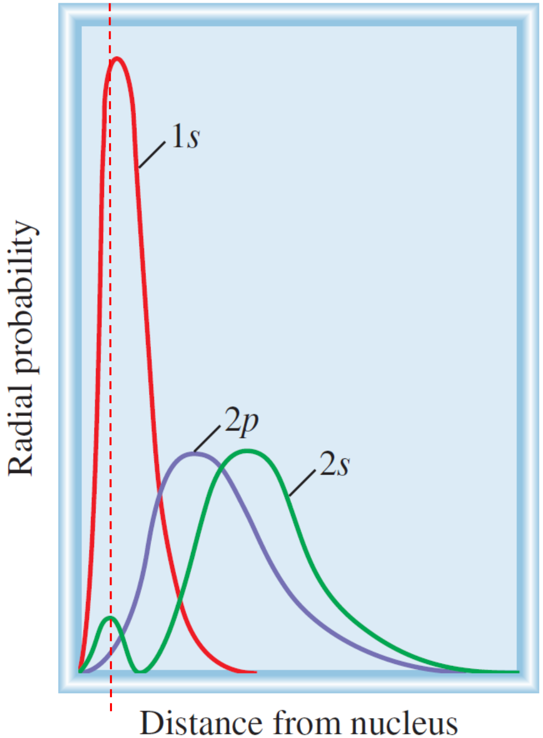
Radial probability plots for the 1s, 2s, and 2p orbitals.

The 1s electrons effectively shield both the 2s and 2p electrons from the nucleus.
The 2s orbital is more penetrating than the 2p orbital.



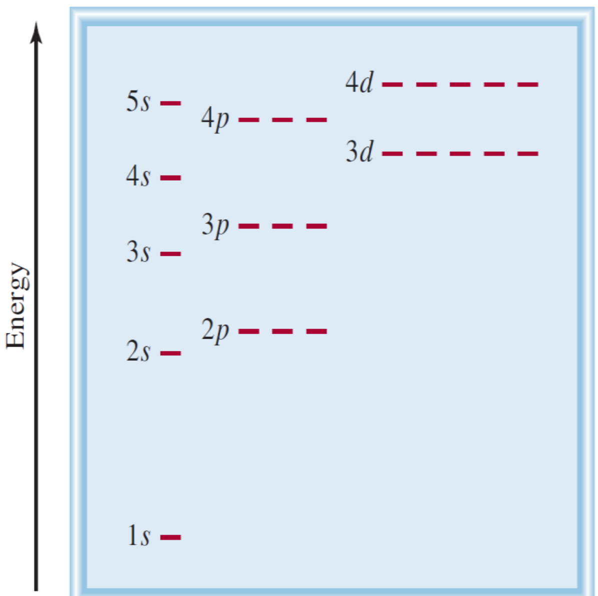
The 2s and 2p orbitals are larger than the 1s orbital, an electron in either of these orbitals will spend more time away from the nucleus than an electron in the 1s orbital.
⇒a 2s or 2p electron being partly "shielded" from the attractive force of the nucleus by the 1s electrons.
⇒The shielding effect is that it reduces the electrostatic attraction between the protons in the nucleus and the electron in the 2s or 2p orbital.

The Shielding Effect in Many-Electron Atoms (1)



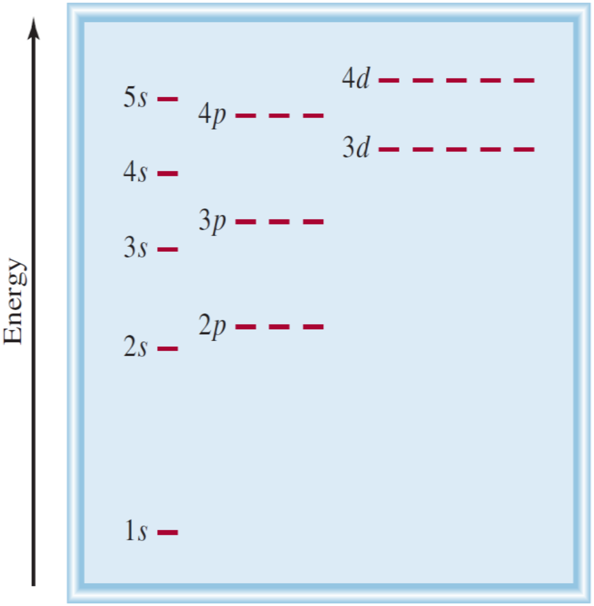
The 2s electron spends most of its time (on average) slightly farther from the nucleus than a 2p electron, the electron density near the nucleus is actually greater for the 2s electron (the small maximum for the 2s orbital).

⇒the 2s orbital is said to be more "penetrating" than the 2p orbital. Therefore, a 2s electron is less shielded by the 1s electrons and is more strongly held by the nucleus.



For the same principal quantum number n, the penetrating power decreases as the angular momentum quantum number ℓ increases, or $s > p > d > f >$·····

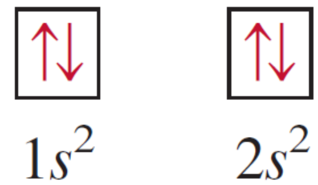
The Shielding Effect in Many-Electron Atoms (2)



<Example> Beryllium atom (Z=4)

electron configuration: $1s^2 2s^2$

orbital diagram: Be

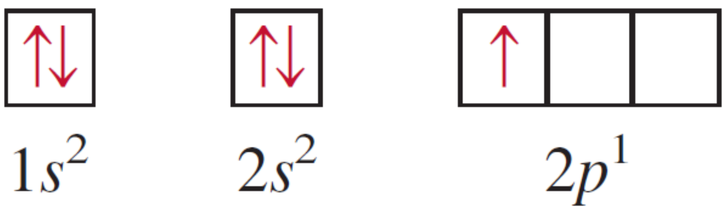


$\Rightarrow$Beryllium is diamagnetic

<Example> Boron atom (Z=5)

electron configuration: $1s^2 2s^2 2p^1$

orbital diagram: B



$\Rightarrow$Boron is paramagnetic

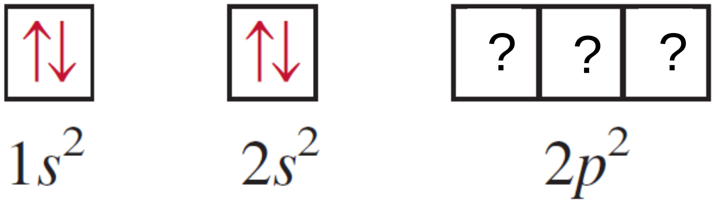
The unpaired electron can be in the $2p_x$, $2p_y$, or $2p_z$ orbital. The choice is completely arbitrary because the three p orbitals are equivalent in energy.

Hund's Rule

<Example> Carbon atom (Z=6)

electron configuration: $1s^2 2s^2 2p^2$

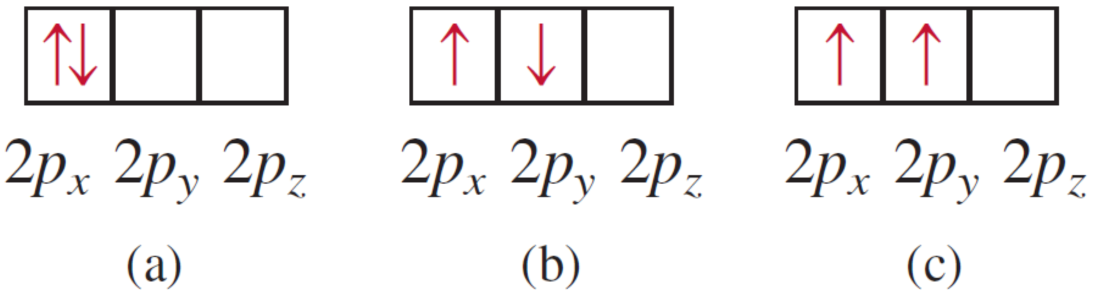
orbital diagram: C



Both (a) and (b) the two spins cancel each other.

In (a), the two electrons are in the same $2p_x$ orbital, and their proximity results in a greater mutual repulsion than when they occupy two separate orbitals, say $2p_x$ and $2p_y$.

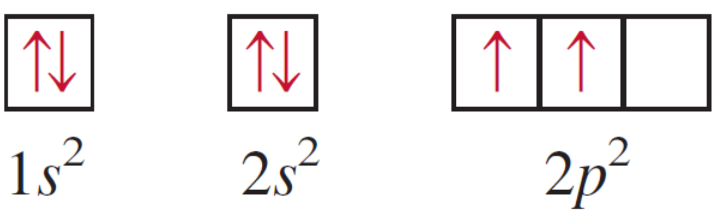
The following are different ways of distributing two electrons among three p orbitals:



None of the three arrangements violates the Pauli exclusion principle: Which one will give the greatest stability?

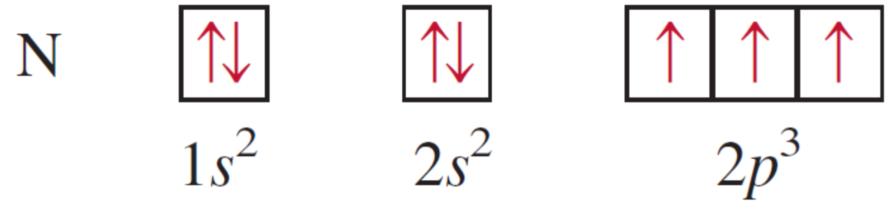
⇒The most stable arrangement of electrons in subshells is the one with the greatest number of parallel spins (***Hund's rule***).

⇒orbital diagram: C

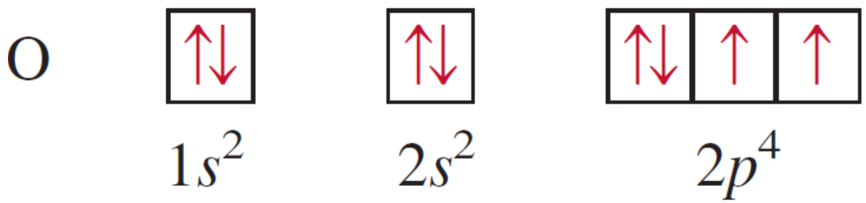


Hund's Rule(1)

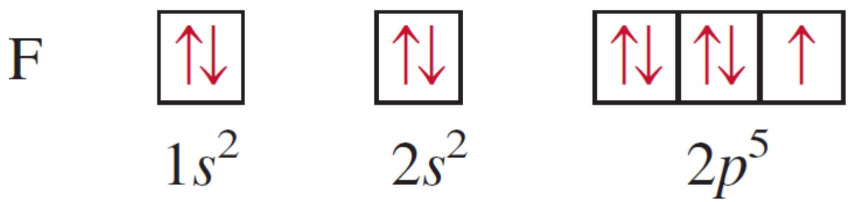
<Example> Nitrogen atom (Z=7)
electron configuration: $1s^2 2s^2 2p^3$

orbital diagram: 

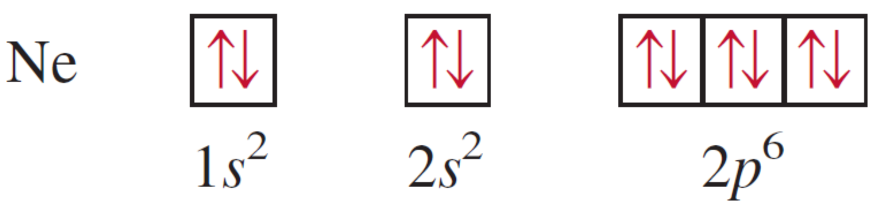
<Example> Oxygen atom (Z=8)
electron configuration: $1s^2 2s^2 2p^4$

orbital diagram: 

<Example> Fluorine atom (Z=9)
electron configuration: $1s^2 2s^2 2p^5$

orbital diagram: 

<Example> Neon atom (Z=10)
electron configuration: $1s^2 2s^2 2p^6$

orbital diagram: 

General Rules for Assigning Electrons to Atomic Orbitals

General rules for determining the maximum number of electrons that can be assigned to the various subshells and orbitals for a given value of n:

1. Each shell or principal level of quantum number n contains n subshells. For example, if $n = 2$, then there are two subshells (two values of ℓ) of angular momentum quantum numbers 0 and 1.

2. Each subshell of quantum number ℓ contains $(2\ell + 1)$ orbitals. For example, if $\ell = 1$, then there are three p orbitals.

3. No more than two electrons can be placed in each orbital. Therefore, the maximum number of electrons is simply twice the number of orbitals that are employed.

4. A quick way to determine the maximum number of electrons that an atom can have in a principal level n is to use the formula $2n^2$.

Example 7.10

What is the maximum number of electrons that can be present in the principal level for which $n = 3$?

Solution When $n = 3$, $\ell = 0, 1$, and 2. The number of orbitals for each value of ℓ is given by

Value of ℓ	Number of Orbitals $(2\ell + 1)$
0	1
1	3
2	5

The total number of orbitals is nine. Because each orbital can accommodate two electrons, the maximum number of electrons that can reside in the orbitals is 2×9, or 18.

Example 7.11

An oxygen atom has a total of eight electrons. Write the four quantum numbers for each of the eight electrons in the ground state.

Solution

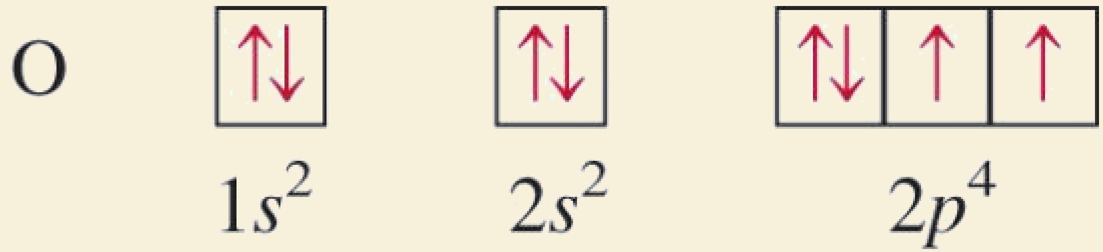
$\because n = 1$, so $\ell = 0$, a subshell corresponding to the 1*s* orbital.

$\Rightarrow$This orbital can accommodate a total of two electrons.

$\because n = 2$, and $\ell = 0$ or 1.

$\Rightarrow$The $\ell = 0$ subshell contains one 2*s* orbital, which can accommodate two electrons.

$\Rightarrow$The remaining four electrons are placed in the $\ell = 1$ subshell, which contains three 2*p* orbitals. The orbital diagram is



Example 7.11 (1)

The results are summarized in the following table:

Electron	n	ℓ	m_ℓ	m_s	Orbital
1	1	0	0	$+\frac{1}{2}$	1*s*
2	1	0	0	$-\frac{1}{2}$	1*s*
3	2	0	0	$+\frac{1}{2}$	2*s*
4	2	0	0	$-\frac{1}{2}$	2*s*
5	2	1	-1	$+\frac{1}{2}$	$2p_x, 2p_y, 2p_z$
6	2	1	0	$+\frac{1}{2}$	$2p_x, 2p_y, 2p_z$
7	2	1	1	$+\frac{1}{2}$	$2p_x, 2p_y, 2p_z$
8	2	1	1	$-\frac{1}{2}$	$2p_x, 2p_y, 2p_z$

The placement of the eighth electron in the orbital labeled $m_\ell = 1$ is completely arbitrary.

It would be equally correct to assign it to $m_\ell = 0$ or $m_\ell = -1$.

First 10 elements has revealed about ground-state electron configurations and the properties of electrons in atoms (1):

1. No two electrons in the same atom can have the same four quantum numbers$\Rightarrow$Pauli exclusion principle.

2. Each orbital can be occupied by a maximum of two electrons. They must have opposite spins, or different electron spin quantum numbers.

3. The most stable arrangement of electrons in a subshell is the one that has the greatest number of parallel spins.$\Rightarrow$Hund's rule.

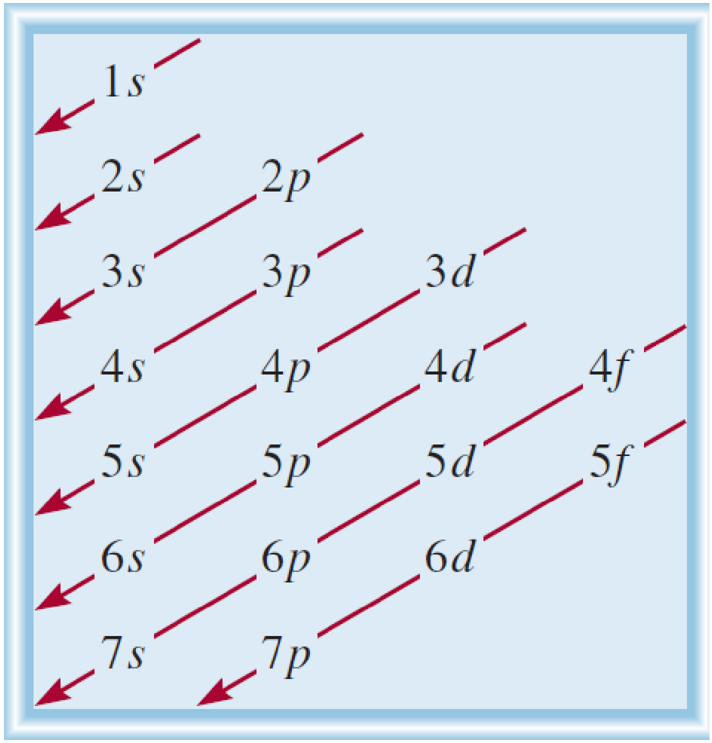
4. Atoms in which one or more electrons are unpaired are paramagnetic. Atoms in which all the electron spins are paired are diamagnetic.

5. In a hydrogen atom, the energy of the electron depends only on its principal quantum number n.

In a many-electron atom, the energy of an electron depends on both n and its angular momentum quantum number ℓ.

First 10 elements has revealed about ground-state electron configurations and the properties of electrons in atoms (2):

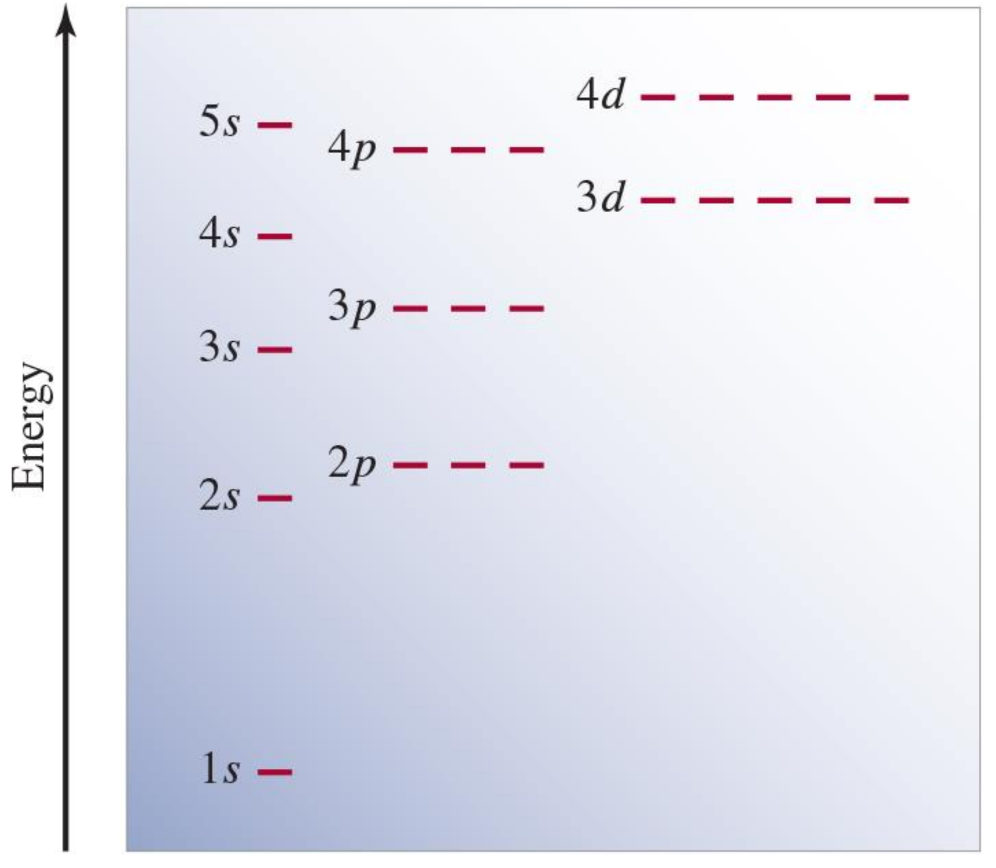
6. In a many-electron atom the subshells are filled in the order as shown in Figure.



7. For electrons of the same principal quantum number, their penetrating power, or proximity to the nucleus, decreases in the order *s* > *p* > *d* > *f*. This means that, for example, more energy is required to separate a 3*s* electron from a many electron atom than is required to remove a 3*p* electron.

The Building-Up Principle: Aufbau Principle

"Fill up" electrons in lowest energy orbitals (***Aufbau principle***)- as protons are added one by one to the nucleus to build up the elements, electrons are similarly added to the atomic orbitals.



Ground State Electron Configurations

Table 7.3 The Ground-State Electron Configurations of the Elements*

Atomic Number	Symbol	Electron Configuration	Atomic Number	Symbol	Electron Configuration	Atomic Number	Symbol	Electron Configuration
1	H	$1s^1$	41	Nb	[Kr]$5s^14d^4$	81		
2	He	$1s^2$	42	Mo	[Kr]$5s^14d^5$	82		
3	Li	[He]$2s^1$	43	Tc	[Kr]$5s^24d^5$	83		
4	Be	[He]$2s^2$	44	Ru	[Kr]$5s^14d^7$	84		
5	B	[He]$2s^22p^1$	45	Rh	[Kr]$5s^14d^8$	85		
6	C	[He]$2s^22p^2$	46	Pd	[Kr]$4d^{10}$	86		
7	N	[He]$2s^22p^3$	47	Ag	[Kr]$5s^14d^{10}$	87		
8	O	[He]$2s^22p^4$	48	Cd	[Kr]$5s^24d^{10}$	88		
9	F	[He]$2s^22p^5$	49	In	[Kr]$5s^24d^{10}5p^1$	89		
10	Ne	[He]$2s^22p^6$	50	Sn	[Kr]$5s^24d^{10}5p^2$	90		
11	Na	[Ne]$3s^1$	51	Sb	[Kr]$5s^24d^{10}5p^3$	91		
12	Mg	[Ne]$3s^2$	52	Te	[Kr]$5s^24d^{10}5p^4$	92		
13	Al	[Ne]$3s^23p^1$	53	I	[Kr]$5s^24d^{10}5p^5$	93		
14	Si	[Ne]$3s^23p^2$	54	Xe	[Kr]$5s^24d^{10}5p^6$	94		
15	P	[Ne]$3s^23p^3$	55	Cs	[Xe]$6s^1$	95		
16	S	[Ne]$3s^23p^4$	56	Ba	[Xe]$6s^2$	96		
17	Cl	[Ne]$3s^23p^5$	57	La	[Xe]$6s^25d^1$	97		
18	Ar	[Ne]$3s^23p^6$	58	Ce	[Xe]$6s^24f^15d^1$	98		
19	K	[Ar]$4s^1$	59	Pr	[Xe]$6s^24f^3$	99		
20	Ca	[Ar]$4s^2$	60	Nd	[Xe]$6s^24f^4$	100	Fm	[Rn]$7s^25f^{12}$
21	Sc	[Ar]$4s^23d^1$	61	Pm	[Xe]$6s^24f^5$	101	Md	[Rn]$7s^25f^{13}$
22	Ti	[Ar]$4s^23d^2$	62	Sm	[Xe]$6s^24f^6$	102	No	[Rn]$7s^25f^{14}$
23	V	[Ar]$4s^23d^3$	63	Eu	[Xe]$6s^24f^7$	103	Lr	[Rn]$7s^25f^{14}6d^1$
24	Cr	[Ar]$4s^13d^5$	64	Gd	[Xe]$6s^24f^75d^1$	104	Rf	[Rn]$7s^25f^{14}6d^2$
25	Mn	[Ar]$4s^23d^5$	65	Tb	[Xe]$6s^24f^9$	105		
26	Fe	[Ar]$4s^23d^6$	66	Dy	[Xe]$6s^24f^{10}$	106		
27	Co	[Ar]$4s^23d^7$	67	Ho	[Xe]$6s^24f^{11}$	107		
28	Ni	[Ar]$4s^23d^8$	68	Er	[Xe]$6s^24f^{12}$	108		
29	Cu	[Ar]$4s^13d^{10}$	69	Tm	[Xe]$6s^24f^{13}$	109		
30	Zn	[Ar]$4s^23d^{10}$	70	Yb	[Xe]$6s^24f^{14}$	110		
31	Ga	[Ar]$4s^23d^{10}4p^1$	71	Lu	[Xe]$6s^24f^{14}5d^1$	111		
32	Ge	[Ar]$4s^23d^{10}4p^2$	72	Hf	[Xe]$6s^24f^{14}5d^2$	112		
33	As	[Ar]$4s^23d^{10}4p^3$	73	Ta	[Xe]$6s^24f^{14}5d^3$	113		
34	Se	[Ar]$4s^23d^{10}4p^4$	74	W	[Xe]$6s^24f^{14}5d^4$	114		
35	Br	[Ar]$4s^23d^{10}4p^5$	75	Re	[Xe]$6s^24f^{14}5d^5$	115		
36	Kr	[Ar]$4s^23d^{10}4p^6$	76	Os	[Xe]$6s^24f^{14}5d^6$	116		
37	Rb	[Kr]$5s^1$	77	Ir	[Xe]$6s^24f^{14}5d^7$	117		
38	Sr	[Kr]$5s^2$	78	Pt	[Xe]$6s^14f^{14}5d^9$	118		
39	Y	[Kr]$5s^24d^1$	79	Au	[Xe]$6s^14f^{14}5d^{10}$			
40	Zr	[Kr]$5s^24d^2$	80	Hg	[Xe]$6s^24f^{14}5d^{10}$			

The electron configurations of all elements except hydrogen and helium are represented by a noble gas core, which shows in brackets the noble gas element that most nearly precedes the element being considered, followed by the symbol for the highest filled subshells in the outermost shells.

[He] helium core: $1s^2$

[Ne] neon core: $1s^22s^22p^6$

[Ar] argon core: [Ne]$3s^23p^6$

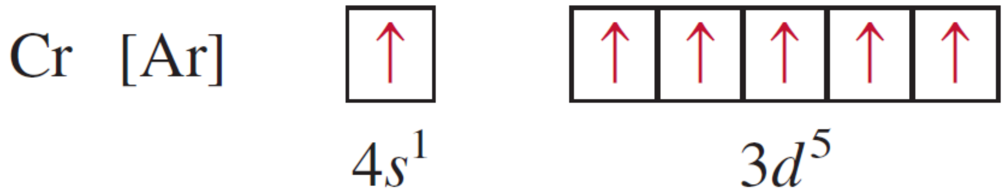
[Kr] krypton core: [Ar]$4s^23d^{10}4p^6$

[Xe] xenon core: [Kr]$5s^24d^{10}5p^6$

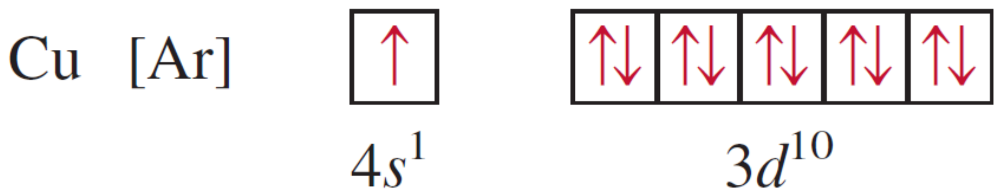
[Rn] radon core: [Xe]$6s^24f^{14}5d^{10}6p^6$

Two irregularities
half-filled and completely filled subshells have extra stability.

<Example> Chromium atom (Z=24)
electron configuration: $[Ar]4s^13d^5$ (not $[Ar]4s^23d^4$)
orbital diagram:



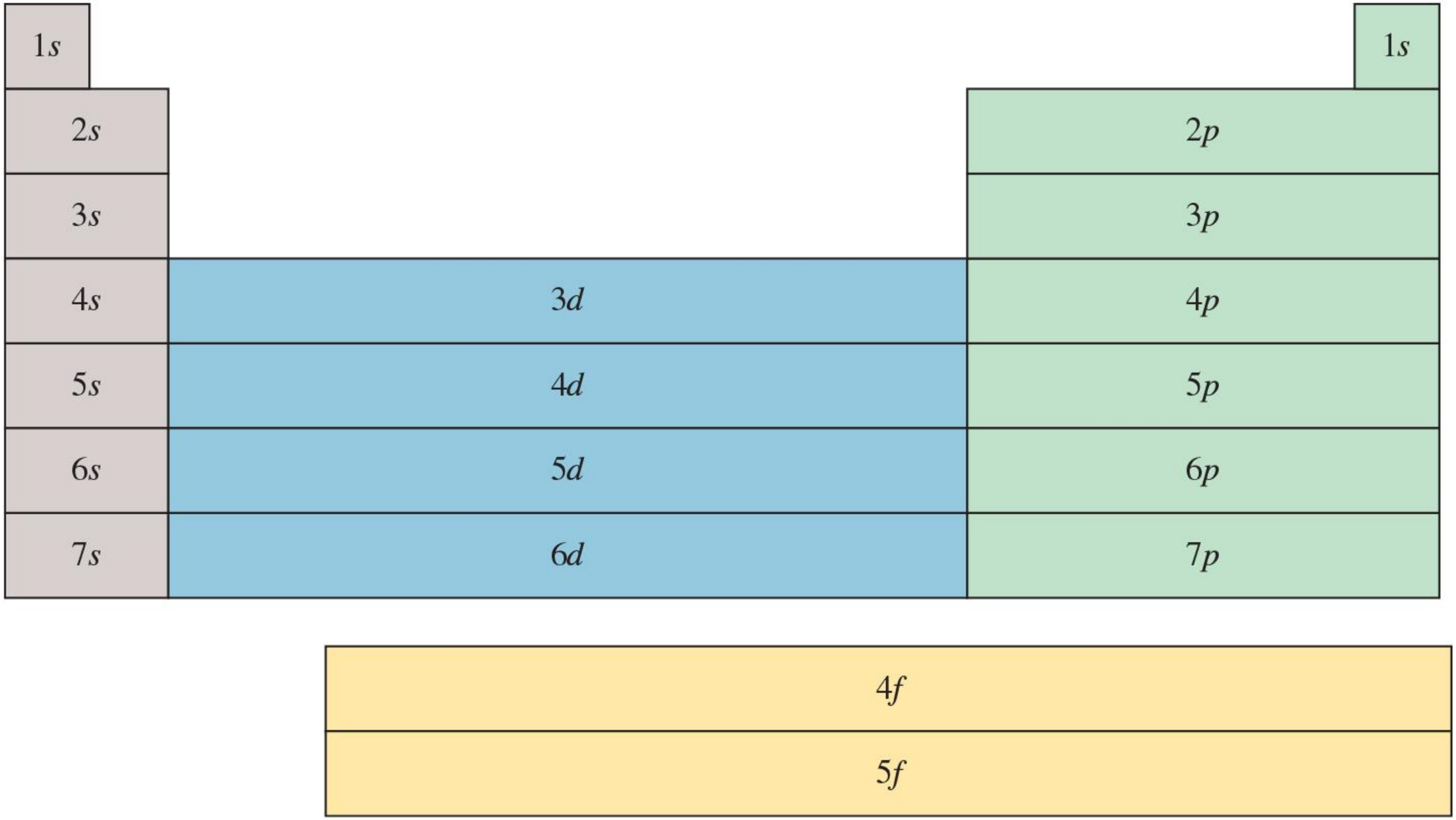
<Example> Copper atom (Z=29)
electron configuration: $[Ar]4s^13d^{10}$ (not $[Ar]4s^23d^9$)
orbital diagram:



The reason for these irregularities is that a slightly greater stability is associated with the half-filled ($3d^5$) and completely filled ($3d^{10}$) subshells. Electrons in the same subshell (in this case, the d orbitals) have equal energy but different spatial distributions. Consequently, their shielding of one another is relatively small, and the electrons are more strongly attracted by the nucleus when they have the $3d^5$ configuration.

Outermost subshell being filled with electrons

Classification of groups of elements in the periodic table according to the type of subshell being filled with electrons.

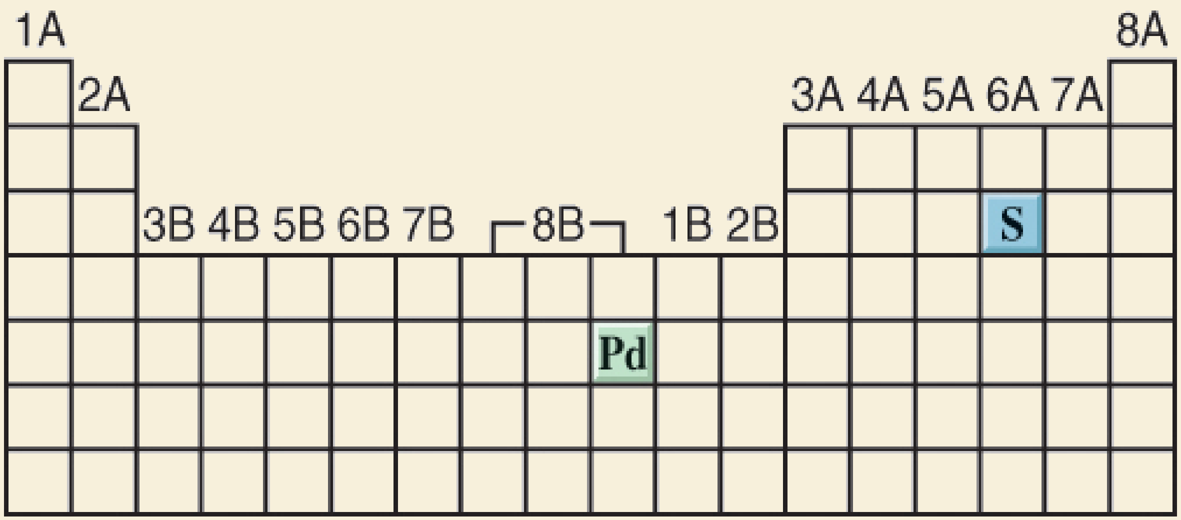


Example 7.12

Write the ground-state electron configurations for

(a) sulfur (S)

(b) palladium (Pd), which is diamagnetic.



Example 7.12 (1)

(a)*Solution* Sulfur has 16 electrons. The noble gas core in this case is [Ne]. [Ne] represents $1s^2 2s^2 2p^6$.
⇒6 electrons to fill the 3*s* subshell and partially fill the 3*p* subshell.
The electron configuration of S is $1s^2 2s^2 2p^6 3s^2 3p^4$ or $[\mathrm{Ne}]3s^2 3p^4$.

(b)*Solution* Palladium has 46 electrons. The noble-gas core in this case is $[\mathrm{Kr}]$. $[\mathrm{Kr}]$ represents $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$
The remaining 10 electrons are distributed among the 4*d* and 5*s* orbitals. The three choices are (1) $4d^{10}$, (2) $4d^9 5s^1$, and (3) $4d^8 5s^2$.
Because palladium is diamagnetic, all the electrons are paired and its electron configuration must be

$$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^{10} \text{ or simply } [\mathrm{Kr}]4d^{10}.$$

The configurations in (2) and (3) both represent paramagnetic elements.

Quantum Dots

Quantum dots are tiny pieces of matter, typically on the order of a few nanometers in diameter, composed of a metal or a semiconductor.

By confining the electrons to such a small volume, the allowed energies of these electrons are quantized. Therefore, if quantum dots are excited to higher energy, only certain wavelengths of light are emitted when the electrons go back to their ground states, just as in the case of the emission spectra of atoms. But unlike atoms, the energy of light omitted from a quantum dot can be “tuned” by varying the size of the quantum dot because that changes the volume within which the electrons are confined.

Emission from dispersed solutions of CdSe quantum dots arranged from left to right in order of increasing diameter (2 nm to 7 nm).

Applications

LEDs (light emitting diodes)

label biological tissue.

quantum computing

photovoltaic cells for harvesting solar energy.

A light micrograph showing the fluorescence of quantum dots in a protozoan, used to study the movement of nanoparticles through the food chain and their potential for accumulation by way of bioconcentration.

7.31 Calculate the wavelength (in nanometers) of a photon emitted by a hydrogen atom when its electron drops from the n=5 state to the n=3 state.

$$\Delta E = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) = (2.18 \times 10^{-18}\ \text{J}) \left(\frac{1}{5^2} - \frac{1}{3^2} \right)$$

$$= -1.55 \times 10^{-19}\ \text{J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34}\ \cancel{\text{J}} \cdot \cancel{\text{s}})(3.00 \times 10^{8}\ \text{m}/\cancel{\text{s}})}{1.55 \times 10^{-19}\ \cancel{\text{J}}}$$

$$= \mathbf{1.28 \times 10^{-6}\ m} = \mathbf{1.28 \times 10^{3}\ nm}$$

7.57 Give the values of the quantum numbers associated with the following orbitals: (a) $2p$, (b) $3s$, (c) $5d$.

(a) $2p$: $n = 2$, $l = 1$, $m_l = 1, 0, \text{or } -1$

(b) $3s$: $n = 3$, $l = 0$, $m_l = 0$ (only allowed value)

(c) $5d$: $n = 5$, $l = 2$, $m_l = 2, 1, 0, -1, \text{or } -2$

7.58 Give the values of the four quantum numbers of an electron in the following orbitals: (a) $3s$, (b) $4p$, (c) $3d$.

(a) $n = 3$. For s orbitals, $\ell = 0$. m_ℓ can have integer values from $-\ell$ to $+\ell$, therefore, $m_\ell = 0$. The electron spin quantum number, m_s, can be either +1/2 or −1/2.

Following the same reasoning as part **(a)**

(b) $4p$: $n = 4$; $l = 1$; $m_l = -1, 0, 1$; $m_s = +1/2, -1/2$

(c) $3d$: $n = 3$; $l = 2$; $m_l = -2, -1, 0, 1, 2$; $m_s = +1/2, -1/2$

7.61 List all the possible subshells and orbitals associated with the principal quantum number n, if n=5.

The allowed values of ℓ =0, 1, 2, 3, and 4. [ℓ= 0 to (n - 1)].

These correspond to the 5s, 5p, 5d, 5f, and 5g subshells.

These subshells each have 1, 3, 5, 7, and 9 orbitals, respectively. (number of orbitals = 2ℓ + 1).

7.92 Which of the following species has the most unpaired electrons? S^+, S, or S^-. Explain how you arrive at your answer.

$\underline{\uparrow\downarrow}$ $\quad$ $\underline{\uparrow}$ $\underline{\uparrow}$ $\underline{\uparrow}$

$3s^2$ $\quad$ $3p^3$

S^+ (5 valence electrons)

3 unpaired electrons

$\underline{\uparrow\downarrow}$ $\quad$ $\underline{\uparrow\downarrow}$ $\underline{\uparrow}$ $\underline{\uparrow}$

$3s^2$ $\quad$ $3p^4$

S (6 valence electrons)

2 unpaired electrons

$\underline{\uparrow\downarrow}$ $\quad$ $\underline{\uparrow\downarrow}$ $\underline{\uparrow\downarrow}$ $\underline{\uparrow}$

$3s^2$ $\quad$ $3p^5$

S^- (7 valence electrons)

1 unpaired electron

S^+ has the most unpaired electrons

7.139

(a) An electron in the ground state of the hydrogen atom moves at an average speed of 5×10^{6} m/s. If the speed is known to an uncertainty of 1 %, what is the uncertainty in knowing its position? Given that the radius of the hydrogen atom in the ground state is 5.29×10^{-11} m, comment on your result. The mass of an electron is 9.1094×10^{-31} kg.

(b) A 3.2-g Ping-Pong ball moving at 50 mph has a momentum of 0.073 kg ? m/s. If the uncertainty in measuring the momentum is 1.0×10^{-7} of the momentum, calculate the uncertainty in the Ping-Pong ball's position.

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

(a)

$$\Delta x \Delta p = \frac{h}{4\pi} \text{ (minimum uncertainty values)} \Rightarrow \Delta x (m \Delta u) = \frac{h}{4\pi}$$

$$\Delta x = \frac{h}{4\pi m \Delta u} = \frac{6.63\times10^{-34}\ \mathrm{J\cdot s}}{4\pi(9.1094\times10^{-31}\ \mathrm{kg})[(0.01)(5\times10^{6}\ \mathrm{m/s})]} = \mathbf{1\times10^{-9}\ m}$$

(b)

$$\Delta x = \frac{h}{4\pi \Delta p} = \frac{6.63\times10^{-34}\ \mathrm{J\cdot s}}{4\pi(1.0\times10^{-7})(0.073\ \mathrm{kg\cdot m/s})} = \mathbf{7.2\times10^{-27}\ m}$$

(e) Superpositions and expectation values

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \xrightarrow{A=B} \psi(x) = A\left(e^{ikx} + e^{-ikx}\right) = 2A\cos kx$$

$$\hat{p}_x\psi = \frac{\hbar}{i}\frac{d\psi}{dx} = \frac{\hbar}{i}\frac{d}{dx}(2A\cos kx) = \frac{\hbar}{i}(-2Ak\sin kx) = -\frac{2k\hbar}{i}\sin kx \Rightarrow \text{not an eigenvalue equation}$$

When the wavefunction of a particle is not an eigenfunction of an operator, the property to which the operator corresponds does not have a definite value.

$$\psi(x) = A\left(e^{ikx} + e^{-ikx}\right) = 2A\cos kx \Rightarrow \underbrace{\psi \text{ is a linear combination of } e^{ikx} \text{ and } e^{-ikx}.}_{\text{the total wavefunction is a superposition of more than one wavefunction.}}$$

$$\Rightarrow \psi = \underbrace{\psi_{\rightarrow}}_{\text{Particle with linear momentum } +k\hbar} + \underbrace{\psi_{\leftarrow}}_{\text{Particle with linear momentum } -k\hbar}$$

The interpretation of this composite wavefunction : if the momentum of the particle is repeatedly measured in a long series of observations, then its **magnitude** will found to be ***kħ*** in all the measurements (because that is the value for each component of the wavefunction). However, because the two component wavefunctions occur equally in the superposition, half the measurements will show that the particle is moving to the right ($p_x = +k\hbar$), and half the measurements will show that it is moving to the left ($p_x = -k\hbar$).

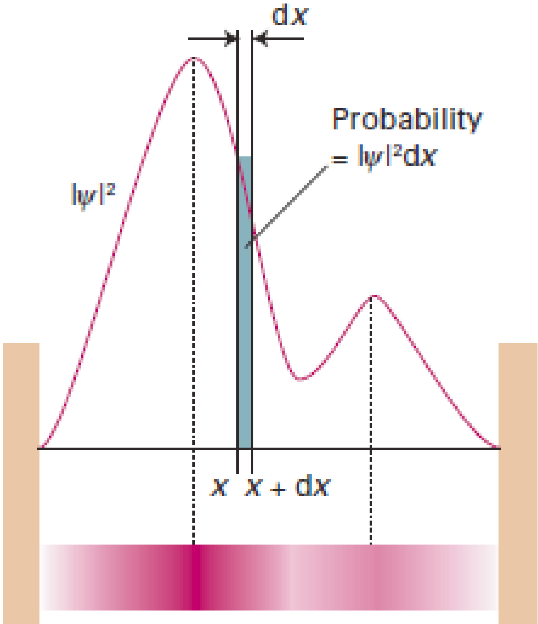


Fig. 7.18 The wavefunction ψ is a probability amplitude in the sense that its square modulus ($\psi^*\psi$ or $|\psi|^2$) is a probability density. The probability of finding a particle in the region dx located at x is proportional to $|\psi|^2$dx. We represent the probability density by the density of shading in the superimposed band.

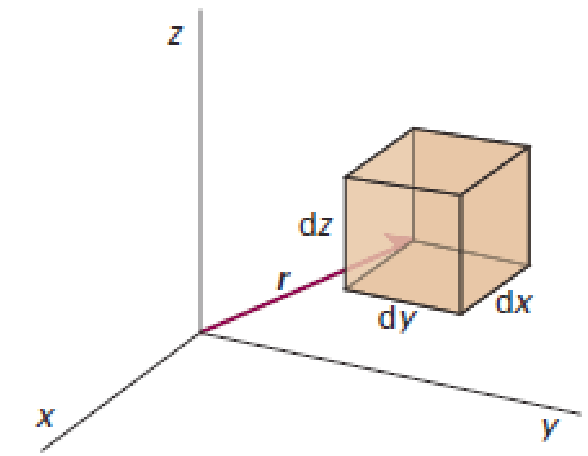


Fig. 7.19 The Born interpretation of the wavefunction in three-dimensional space implies that the probability of finding the particle in the volume element $\mathrm{d}\tau = \mathrm{d}x\mathrm{d}y\mathrm{d}z$ at some location $\boldsymbol{r}$ is proportional to the product of dτ and the value of $|\psi|^2$ at that location.

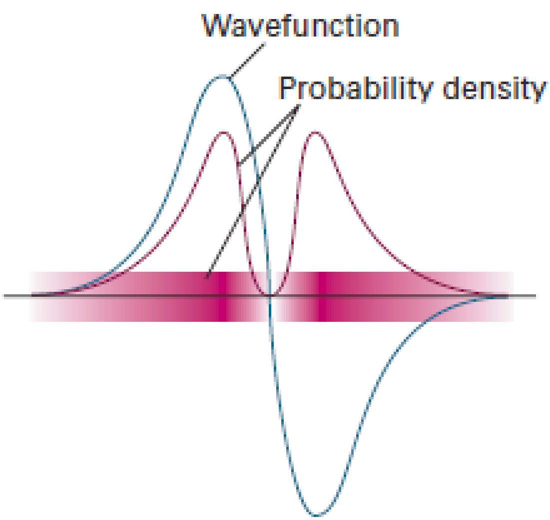


Fig. 7.20 The sign of a wavefunction has no direct physical significance: the positive and negative regions of this wavefunction both correspond to the same probability distribution (as given by the square modulus of ψ and depicted by the density of shading).